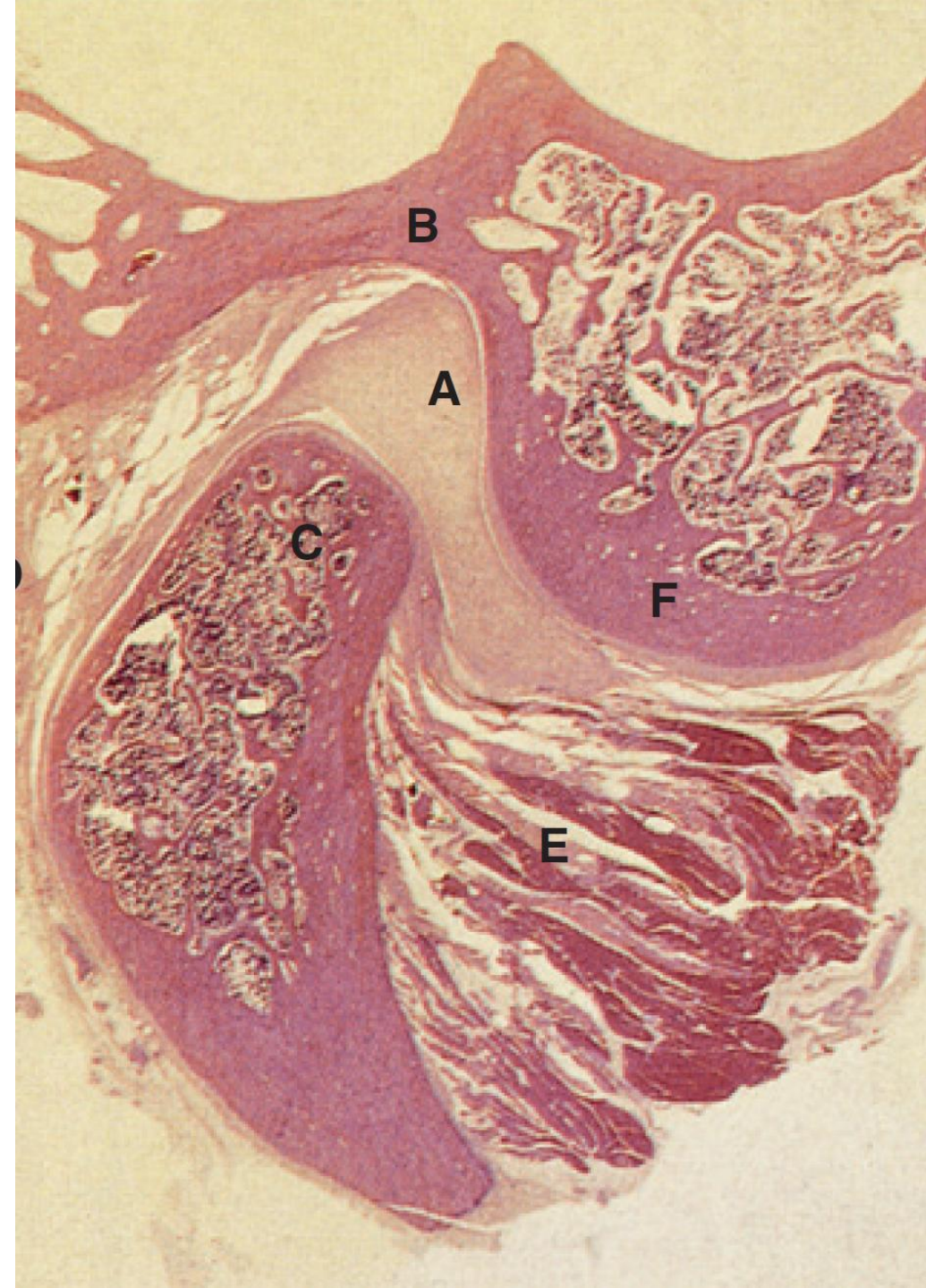


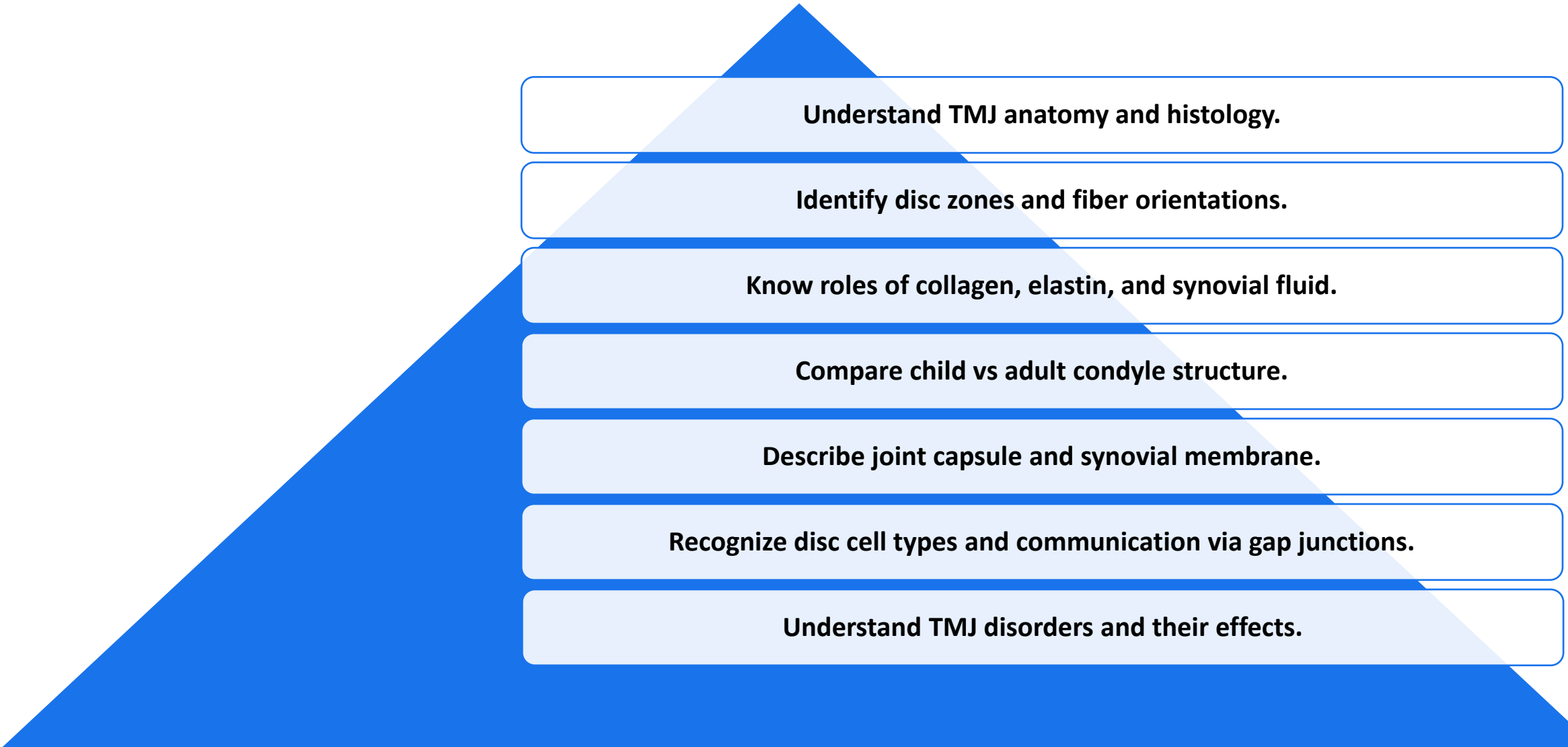
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# Histology of Temporomandibular Joint

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# Learning Objectives – TMJ



**Understand TMJ anatomy and histology.**

**Identify disc zones and fiber orientations.**

**Know roles of collagen, elastin, and synovial fluid.**

**Compare child vs adult condyle structure.**

**Describe joint capsule and synovial membrane.**

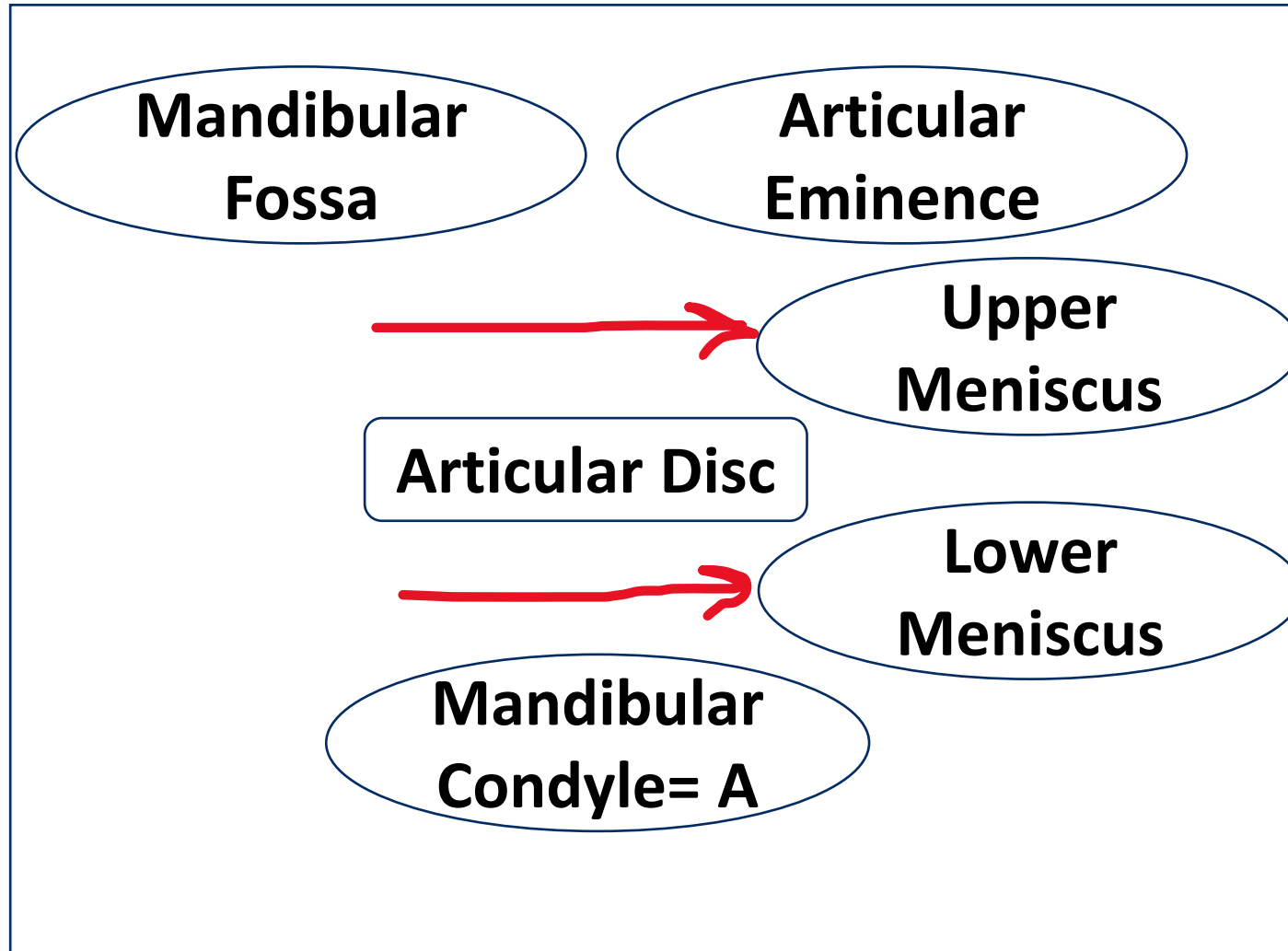
**Recognize disc cell types and communication via gap junctions.**

**Understand TMJ disorders and their effects.**



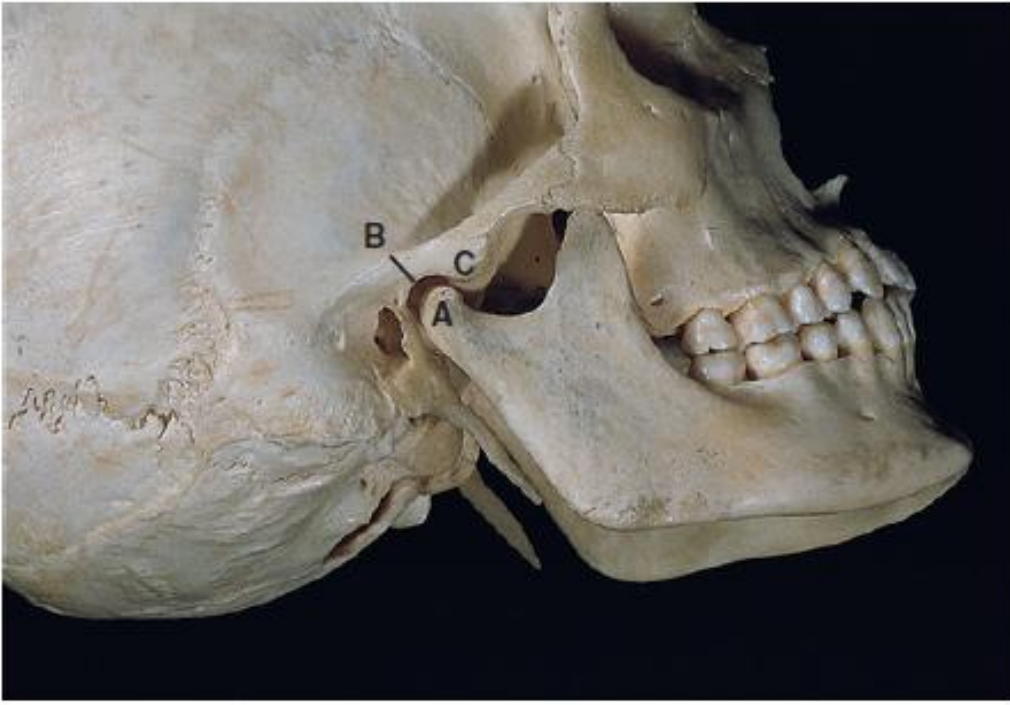
- Think of the TMJ as a hybrid joint —
  - ✓ Inferior compartment: hinge (rotation)
  - ✓ Superior compartment: sliding door (translation).
- It's the only joint in the body where both sides must move in perfect synchrony.
- Bite the wrong way for years, and the whole system breaks down.





**Why should you care?** TMJ disorders affect ~12% of the population. Your future patients will come in with jaw pain, clicking, and headaches — and it starts with understanding this histology.





**Fig. 3.1** The osteology of the temporomandibular joint. A = mandibular condyle; B = mandibular fossa of temporal bone; C = articular eminence of temporal bone.

*FOSSA = REST*  
*EMINENCE = FUNCTION.*

Three bony landmarks you must know:

- ✓ **Mandibular condyle (A)** — the moving ball.
  - Convex, covered in fibrocartilage.
  - The growth centre in children.
- ✓ **Mandibular fossa (B)** — the socket (concave).
  - Thin fibrous covering.
  - Posteriorly non-weight-bearing.
- ✓ **Articular eminence (C)** — the slope in front of the fossa.
  - The true load-bearing surface.
  - Thicker fibrous covering here.

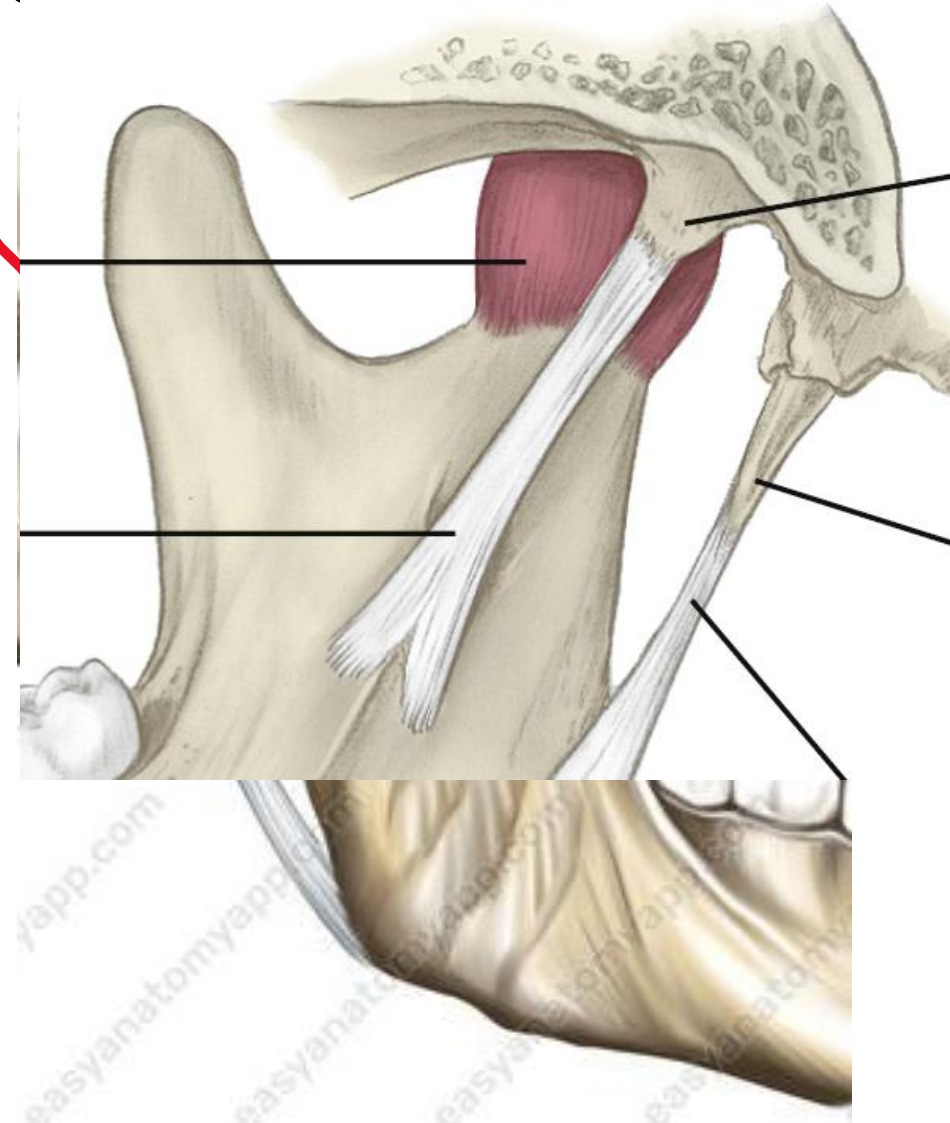
## QUICK RECALL :

1. Which part of the temporal bone is the true weight-bearing articular surface?
2. What type of cartilage covers the TMJ articular surfaces? Why is this unique?

# The joint capsule — a two-layered overcoat

## 1. Outer fibrous layer:

- **Dense fibrous connective tissue** (posterior to tubercle), reinforced by the **lateral TMJ ligament** (anterior to tubercle).
- Resists dislocation. Provides **mechanical support and stability**
- Superior attachment: Around the **articular eminence** and **articular/ mandibular fossa** of the temporal bone.
- Inferior attachment: around the neck of condyle.



# The joint capsule — a two-layered overcoat

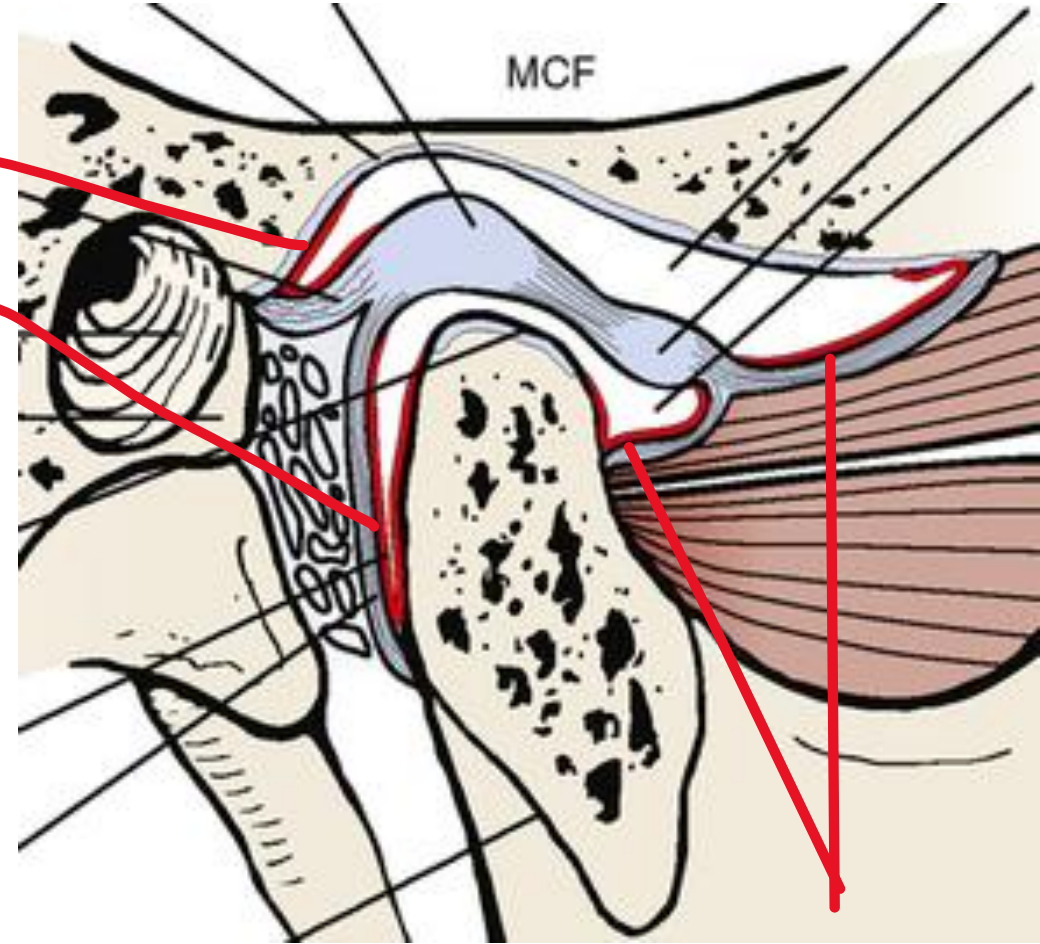
## 2. Inner synovial membrane (Synovium):

- Thin, inner vascular connective tissue.
- Does **NOT** line articular surfaces or disc.
- Secretes synovial fluid → lubricates the joint and nourishes the **avascular disc**.

### Synovial fluid:

**Inferior cavity:** ~1 mL

**Superior cavity:** Slightly more





# Synovial membrane

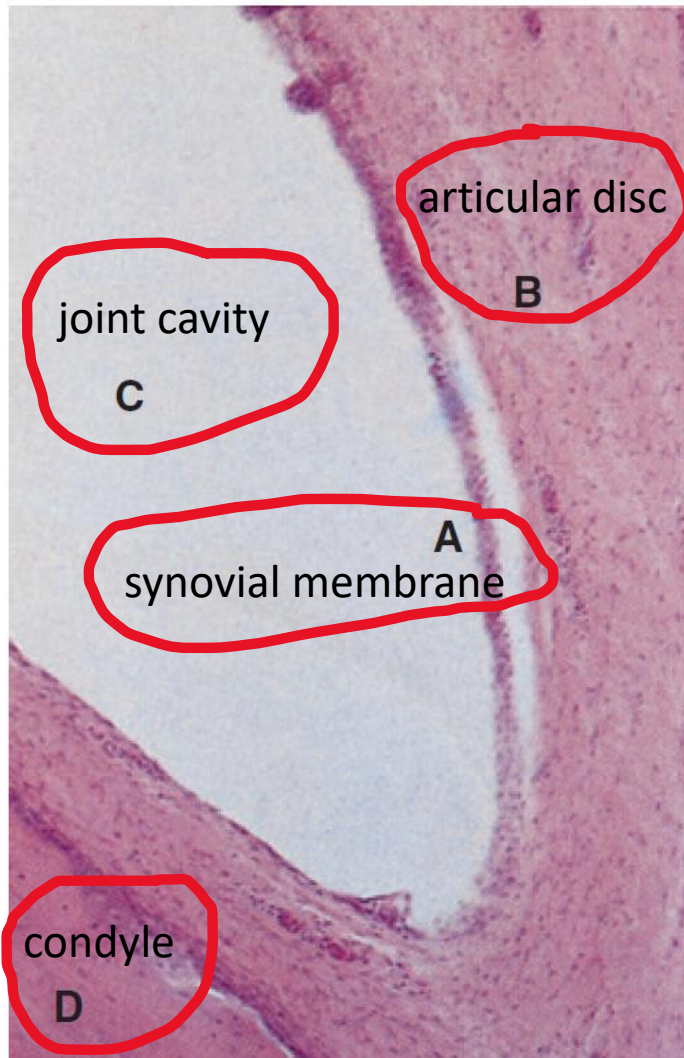


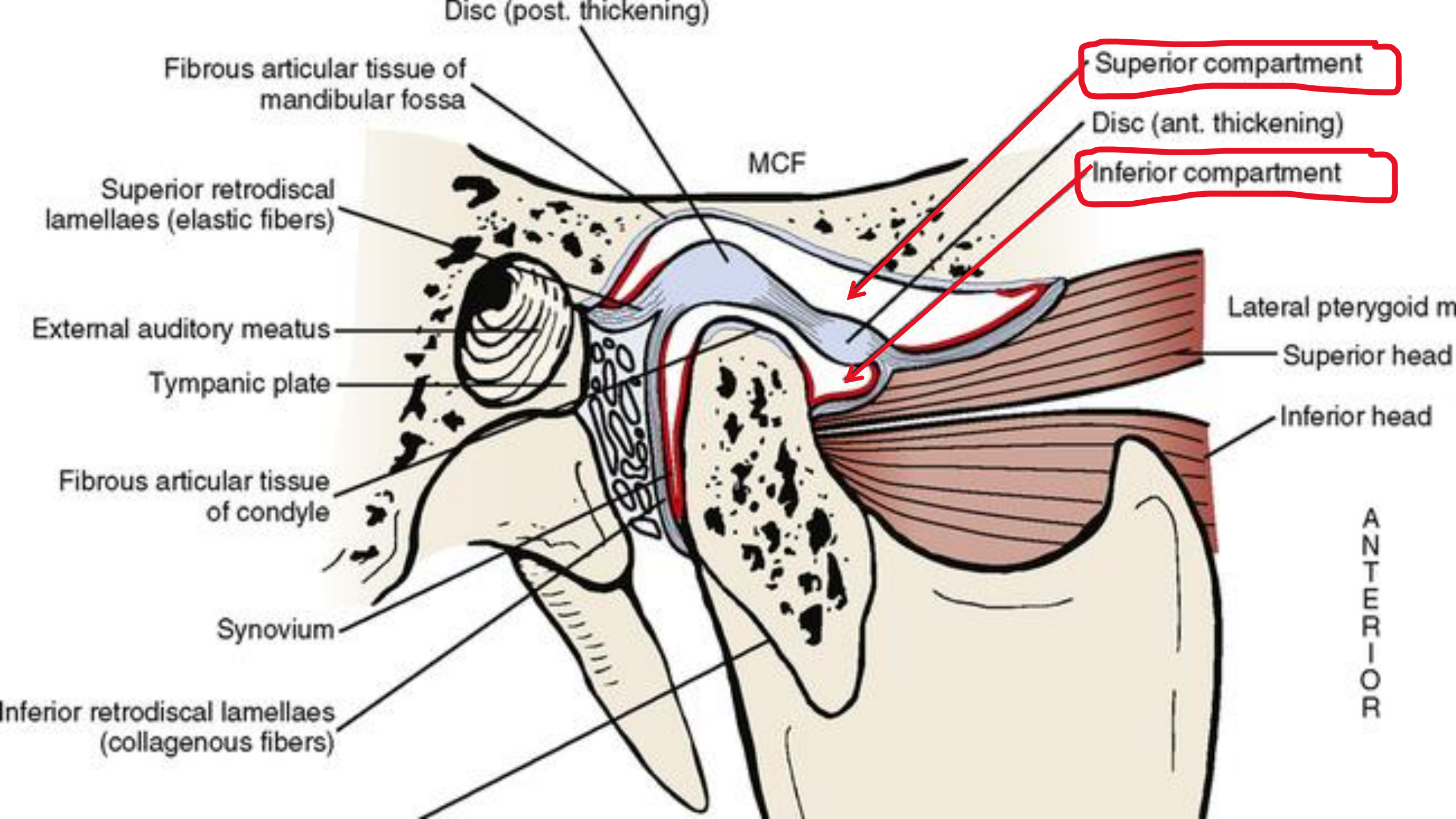
Fig. 15.20

**Type A cells**  
Macrophage-like.  
Phagocytic — clean up debris in the joint cavity.

**Type B cells**  
Fibroblast-like. Secrete hyaluronate (the main lubricant in synovial fluid).

**With age**  
Synovial folds (villi) increase in size and number. More visible in elderly patients.

**Active:** High metabolism, regenerative; **folded at rest**, flattens with movement.

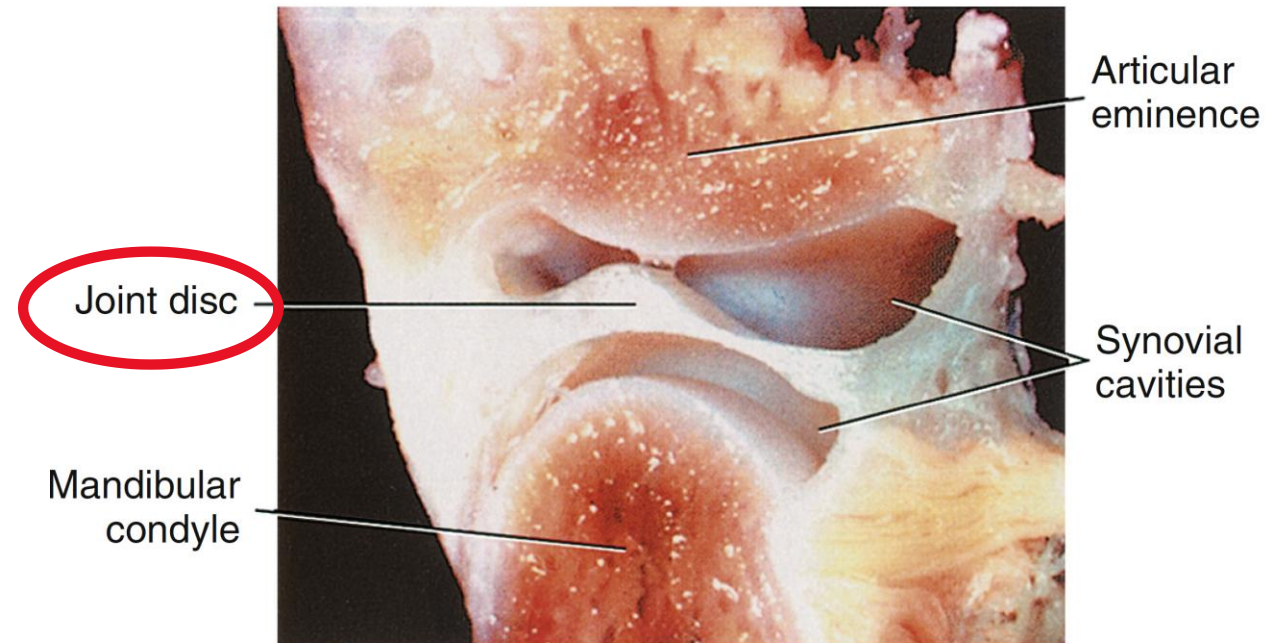
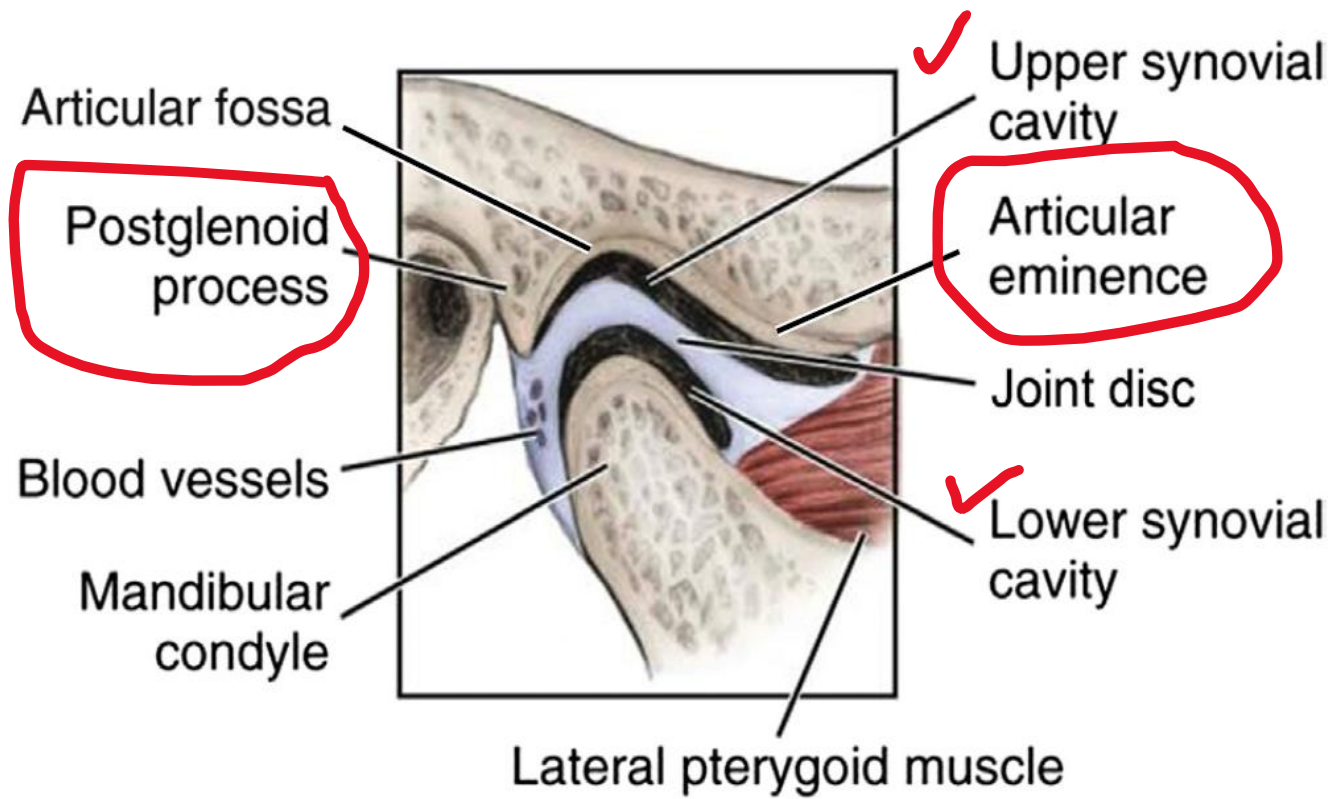


# Synovial Fluid

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- **Secreted by:** Synovial membrane; fills joint cavities.
- **Pressure:** Subatmospheric at rest, elevated during mastication and in bruxism.
- **Functions:** Lubrication, Nutrition for avascular disc & articular surfaces
- **Key Components:**  
Water (major), Hyaluronan → main lubricant, Lubricin (proteoglycan 4), Phospholipids, Bioactive molecules (e.g., TGF- $\beta$ , VEGF, MMPs, TIMPs, RANKL, ROS, IgA, IgG)





**FIGURE 19-2** Block dissection of the temporomandibular joint. (From Nanci A: *Ten Cate's oral histology*, ed 8, St Louis, 2013, Mosby/Elsevier.)

# Synovial Fluid Functions



## **Lubrication**

Reduces friction between disc, condyle, and fossa. Normal TMJ has very low friction coefficient ( $\sim 0.002$ ).

## **Nutrition**

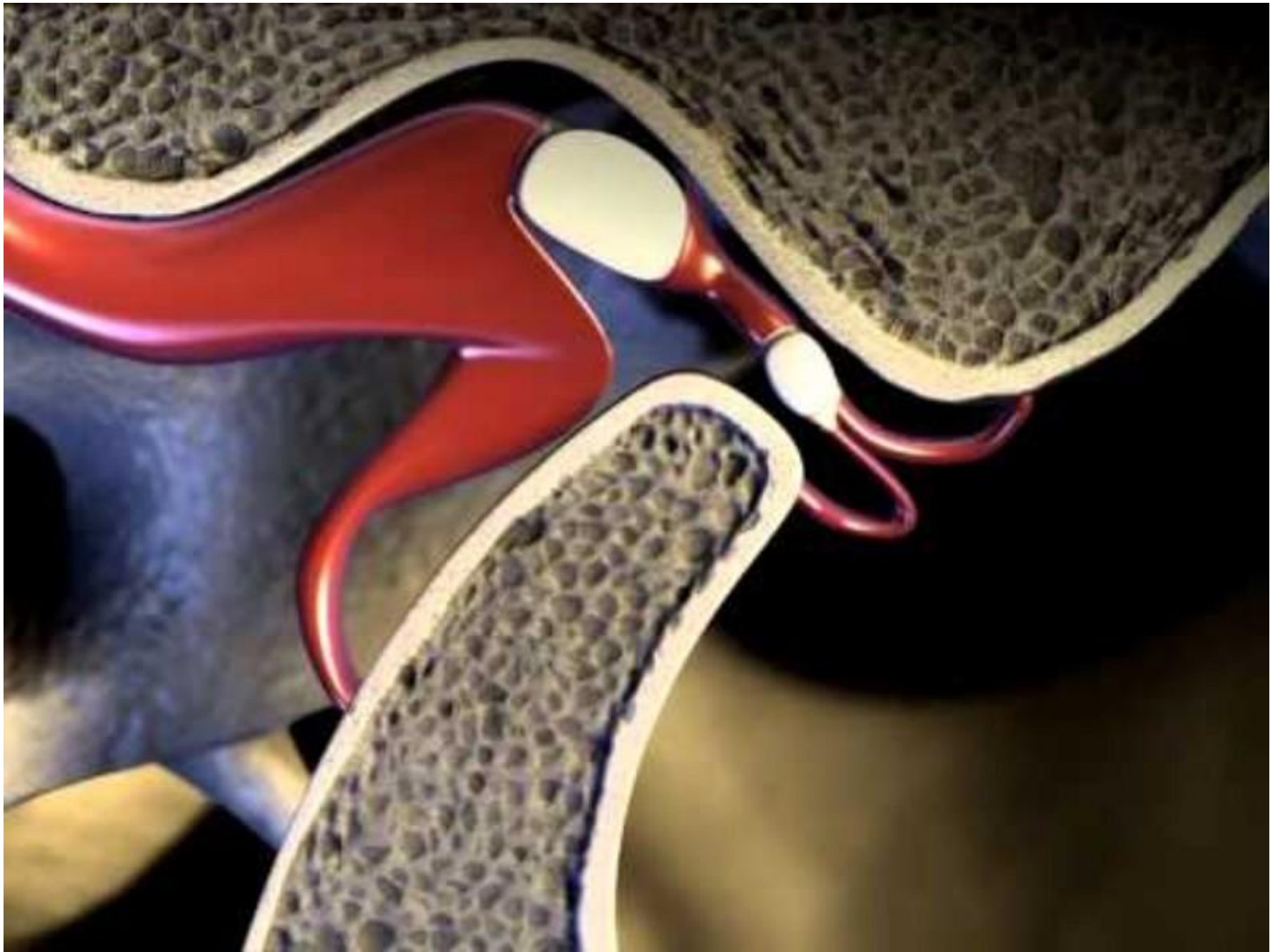
The disc and articular surfaces are avascular — synovial fluid is their only nutrient source. Diffusion + imbibition driven by joint movement.

## **Immune surveillance**

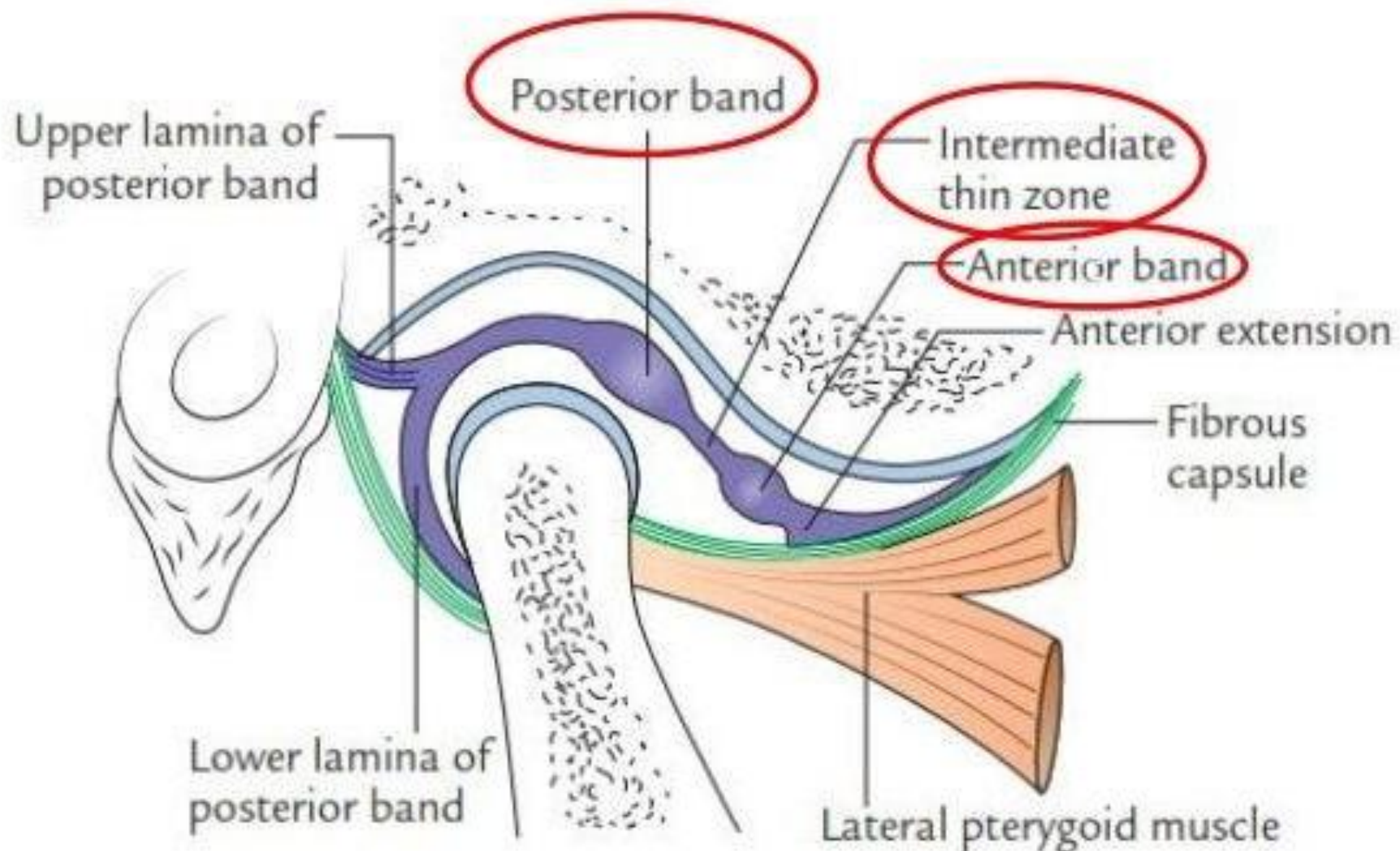
Contains IgA, IgG, and macrophage-like Type A cells. First line of defense against intra-articular infections.



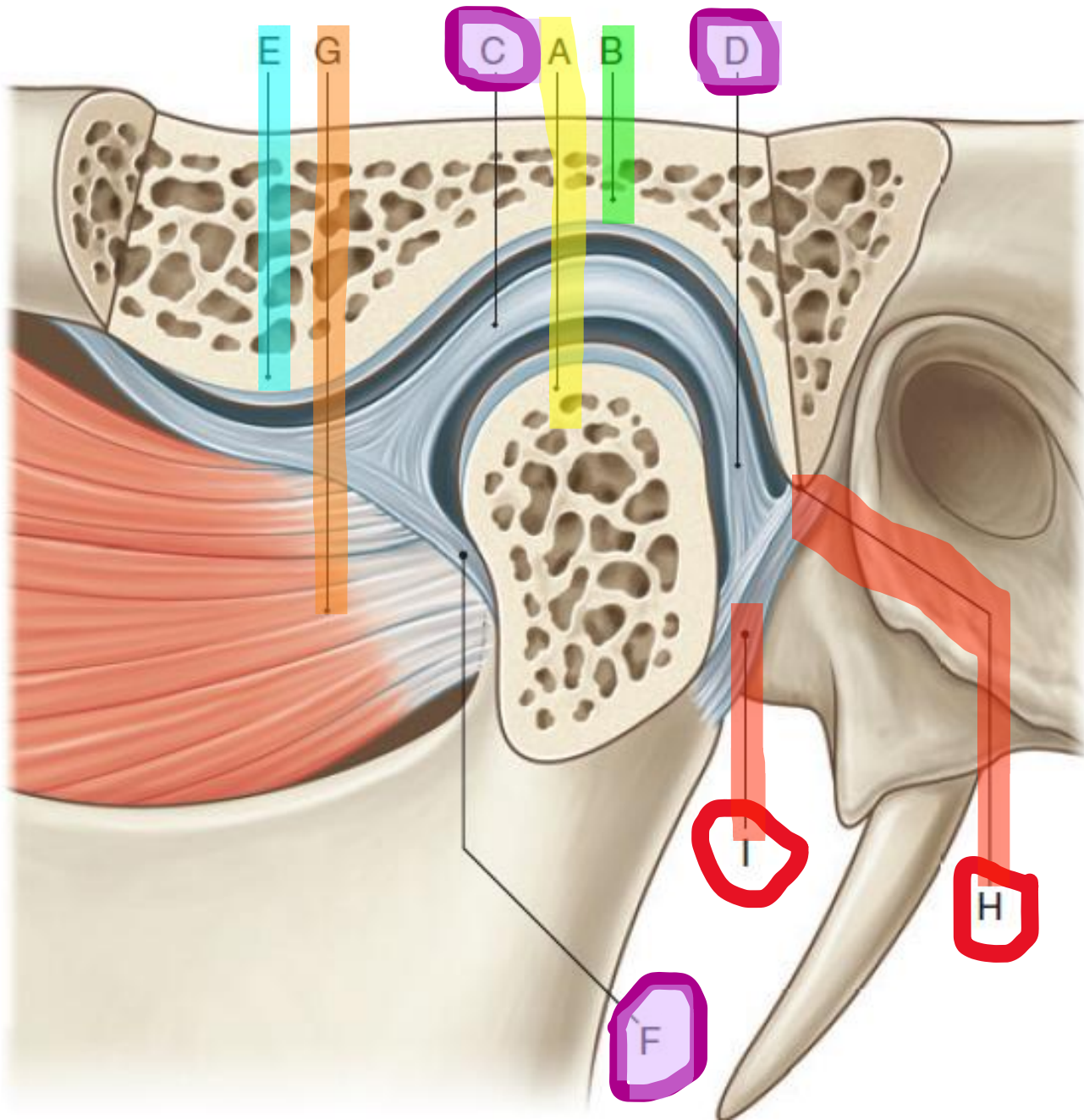




## Articular surfaces and disc







**Fig. 3.8 Diagrammatic representation of the intra-articular disc.**

A = mandibular condyle;

B = mandibular fossa;

C = central part of intra-articular disc;

D = posterior part of intra-articular disc;

E = articular eminence;

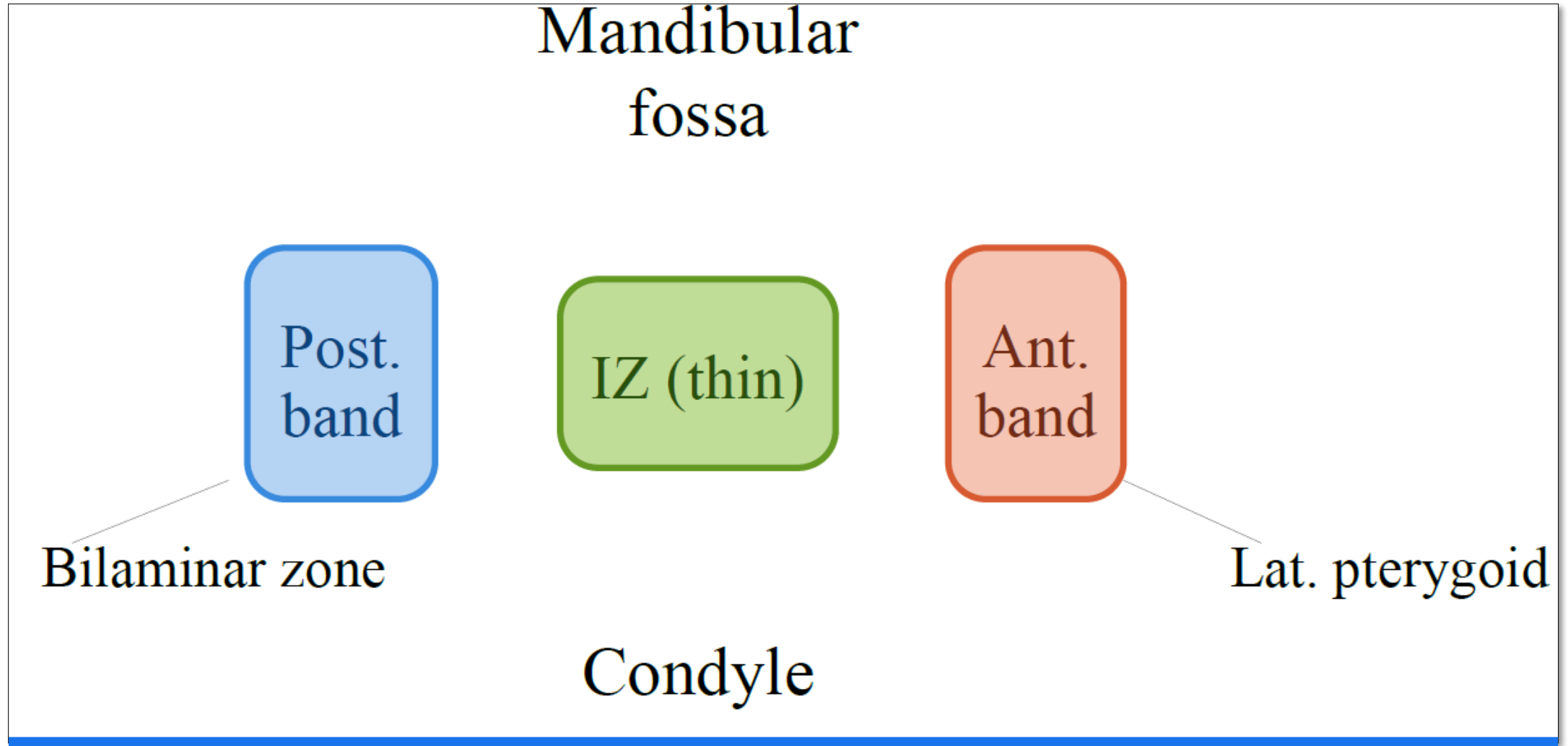
F = attachment of anterior part of intra-articular disc to anterior margin of mandibular condyle;

G = lateral pterygoid muscle;

H = attachment of upper part of bilaminar zone;

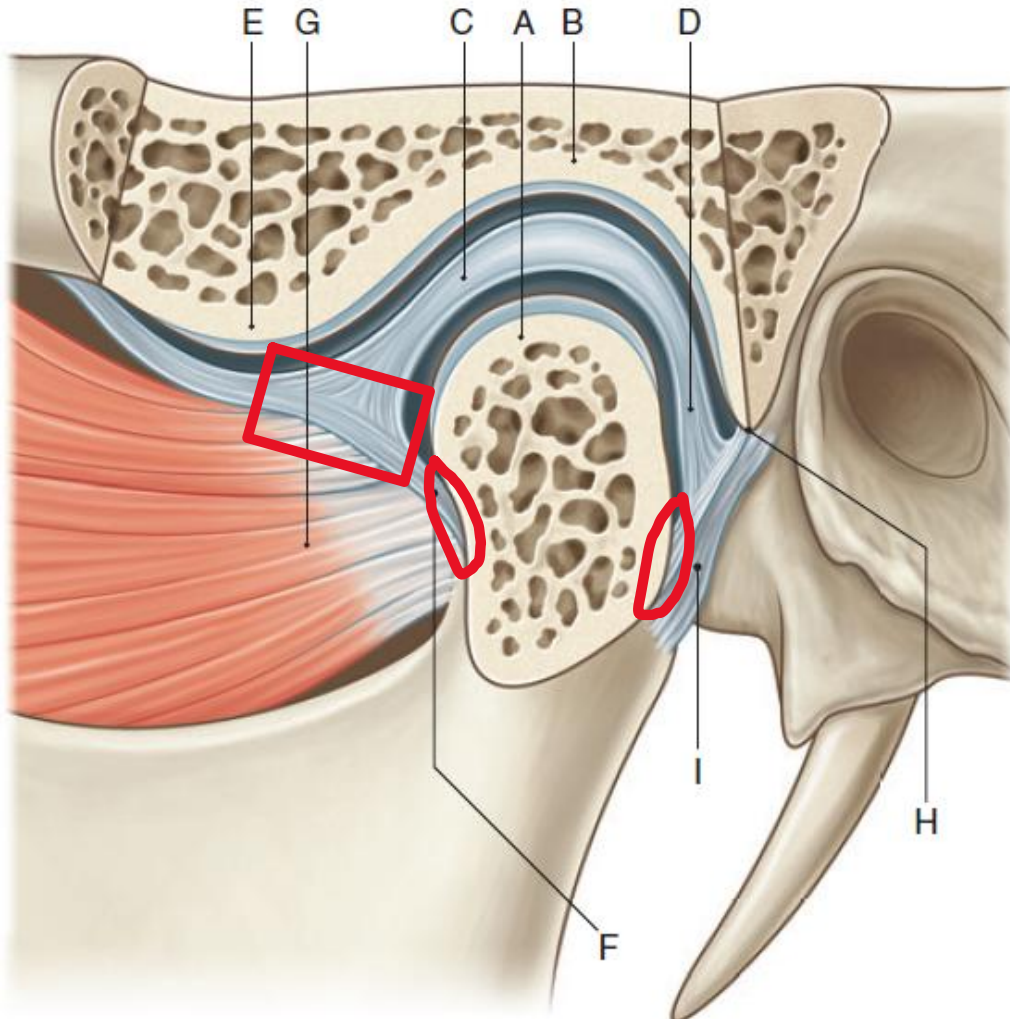
I = attachment of the lower part of bilaminar zone.

# The articular disc — structure, zones, and attachments





# Attachments of the TMJ Articular Disc



## Lateral & Medial Attachments

- The disc is firmly attached to the lateral and medial poles of the mandibular condyle to move with it.

## Anterior Attachment

- Indirectly via the joint capsule.

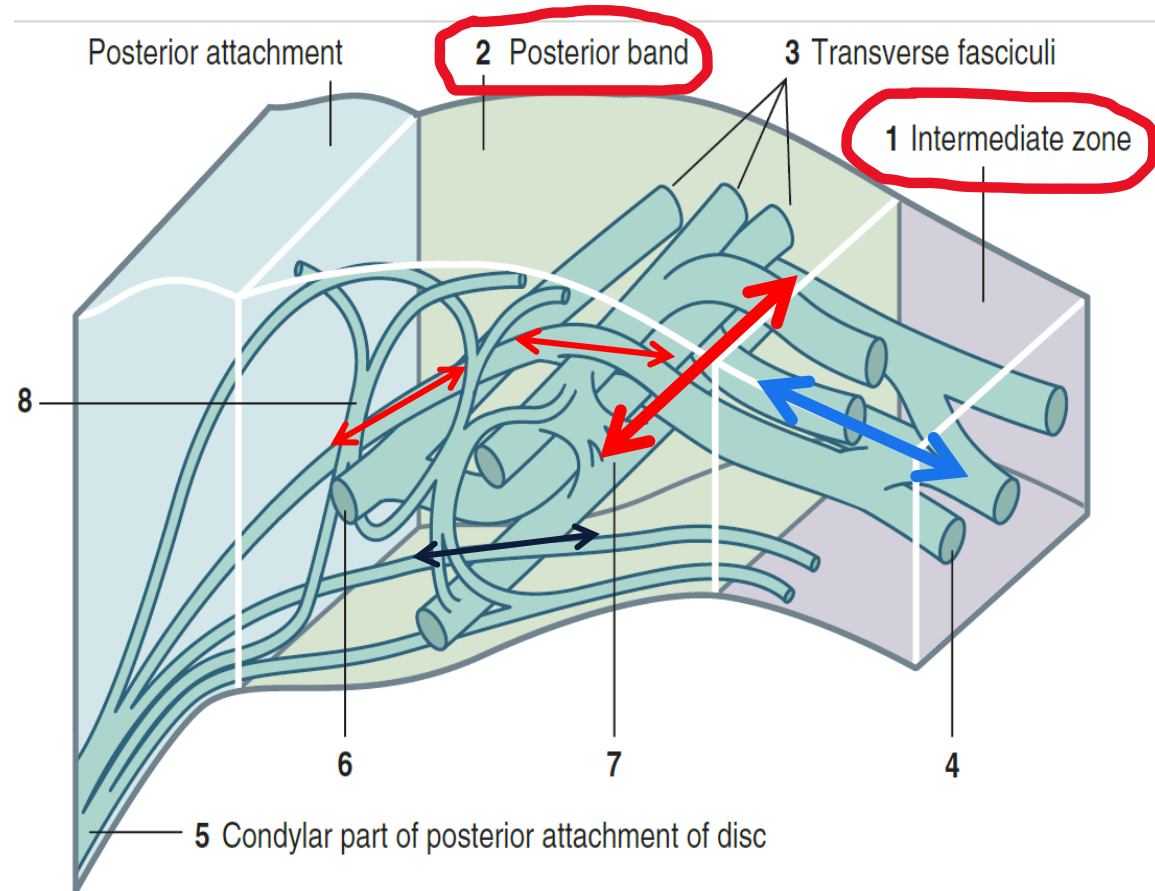
# Attachments of the TMJ Articular Disc



## Posterior Attachments

- Upper Posterior Lamina: Attached to the postglenoid process of the temporal bone.
- Lower Posterior Lamina: Attached to the neck of the mandibular condyle.

# Collagen Fiber Arrangement the intra-articular disc



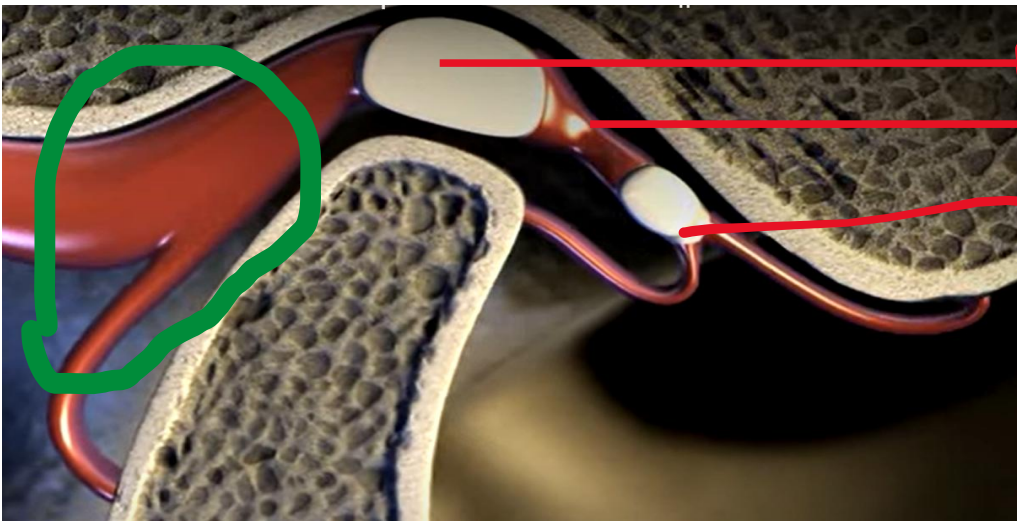
- Composed mainly of **type I collagen** (~80% of dry weight);
- **Traces** of types II, III, VI, IX, XII.
- **Intermediate zone:** fibers run anteroposteriorly.
- **Anterior/posterior bands:** fibers run transversely, vertically, and obliquely.
- **Peripheral fibers:** arranged circumferentially.
- **Fiber bundles** branch, decussate, and join.
- **Crimping seen in superficial fibers** → indicates compressional and tensional loading.

# Biochemical composition of the intra-articular disc

## Box 15.1 **Composition of intra-articular disc**

- 1) Water: 65–80%
- 2) Macromolecules: 20–35%
  - Collagen: 70–85%
  - Cells: 10%
  - Extracellular matrix components: 10% – e.g., proteoglycans, glycoproteins (fibronectin), glycosaminoglycans (chondroitin sulphate), elastin





## Biochemical composition of the intra-articular disc

### ELASTIN FIBRES

- ~5% of dry weight;
- decrease with age.
- Prominent in upper bilaminar zone, more lateral than medial.
- Present in anterior/posterior bands,
- absent in intermediate zone.
- Function: Disc recoil after loading / displacement.

### GROUND SUBSTANCE

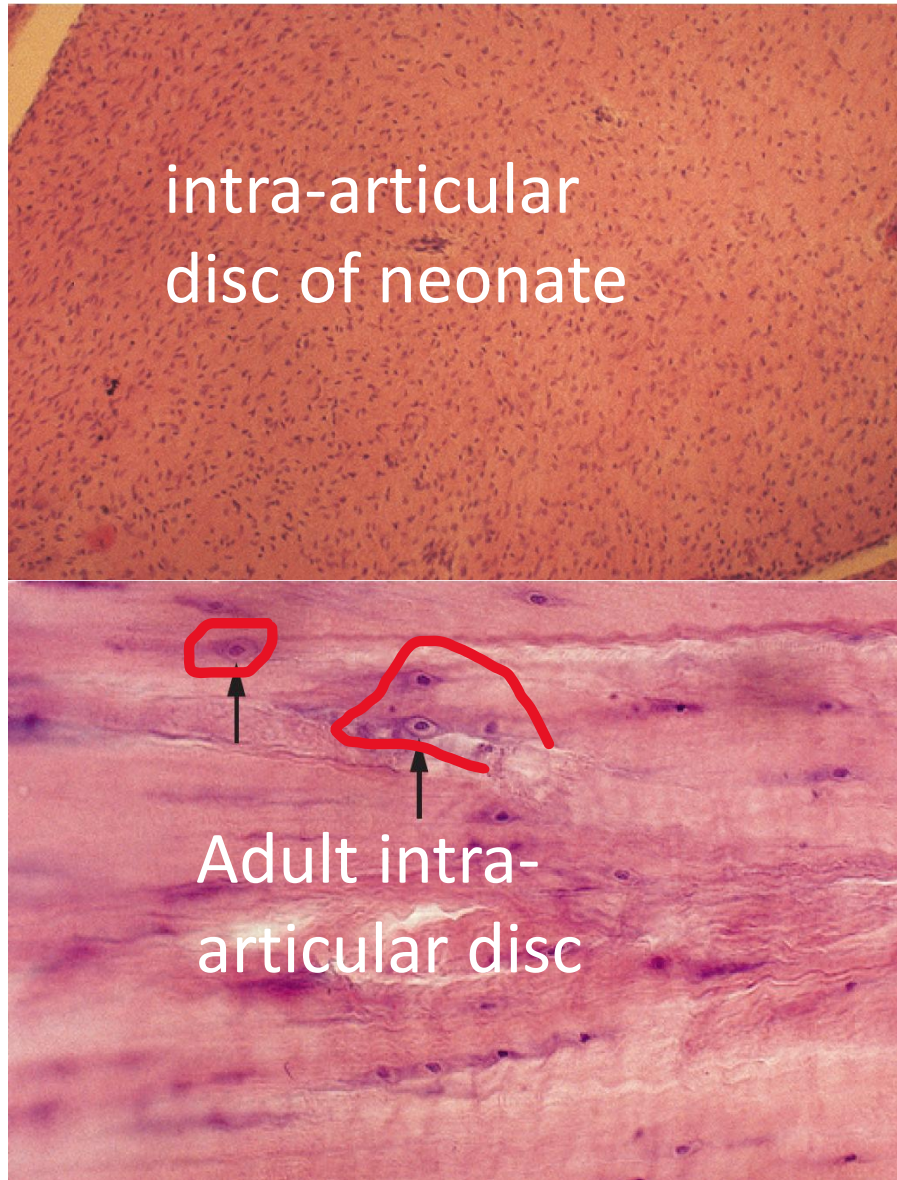
- ~5% of dry weight;
- richest in intermediate zone, more medially.
- Main GAGs: chondroitin sulfate, dermatan sulfate → absorb water, resist compression.
- Forms large proteoglycans (e.g., aggrecan-like).
- Fibronectin & tenascin also present.

# Biochemical composition of the intra-articular disc

| Zone                     | Main Molecules                              | Function                                        |
|--------------------------|---------------------------------------------|-------------------------------------------------|
| <b>Intermediate zone</b> | Collagen ↑,<br>Proteoglycan ↓               | Strength +<br>load-bearing                      |
| <b>Posterior zone</b>    | Proteoglycan ↑,<br>GAG ↑,<br>Glycoprotein ↑ | Shock<br>absorption +<br>elasticity +<br>repair |

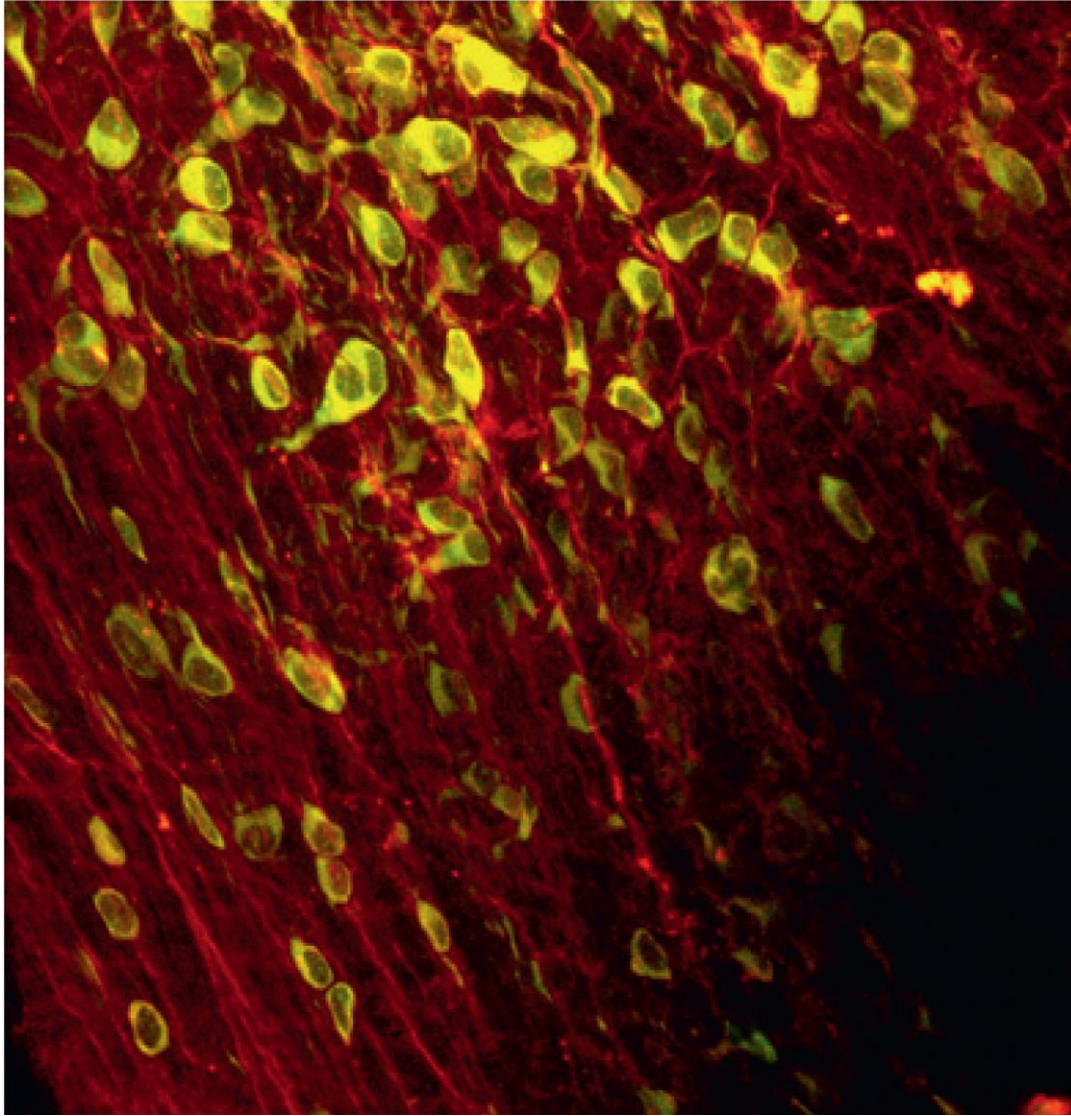
- **Intermediate Zone** =  
Strong (collagen)
- **Posterior Zone** =  
Soft & Vascular  
(proteoglycan + GAG +  
glycoprotein)

# Cells of the TMJ Intra-articular Disc



- More **numerous at birth**, become **sparse in adults**.
- Cell types: Flattened (fibroblast-like) and rounded (chondrocyte-like).
- Contain moderate organelles → ECM synthesis;
- Pericellular matrix mostly absent, seen only in older animals (no capsule).
- Long cytoplasmic processes up to **100  $\mu\text{m}$**  — contain Connexin 43 (gap junction protein)





## Confocal Image of TMJ Disc Cells (×250)

- **Green (Phalloidin):** Highlights cell bodies and long, fine cytoplasmic processes extend up to 100  $\mu\text{m}$ . (F-actin).
- **Red (Connexin 43 antibody):** Labels gap junction protein within the cell processes.
- These processes likely facilitate diffusion of nutrients and fluids from peripheral blood vessels to the avascular centre (IZ) of the disc.



# Vascularization of Intra Articular Disc



- ❖ **Birth:** Vascular → **Adult:** Avascular, vessels at periphery.
- **Superior lamella (posterior):** Vascular, venous engorgement during jaw opening (4–5× volume).
  - ✓ Linked to: Pterygoid venous plexus.
  - ✓ Jaw closure: Blood drains with elastic recoil
- **Inferior lamella (posterior):** Avascular, inelastic.

# Intra Articular Disc Vascularity-Summary

| Region               | Vascularity     |                                          |
|----------------------|-----------------|------------------------------------------|
| IZ (centre)          | Avascular       | Load-bearing — vessels would be crushed  |
| Periphery            | vascular        | Nutrient supply to outer disc            |
| Upper bilaminar zone | Richly vascular | Venous engorgement on jaw opening (4–5×) |
| Lower bilaminar zone | Avascular       | Inelastic — mainly collagen              |

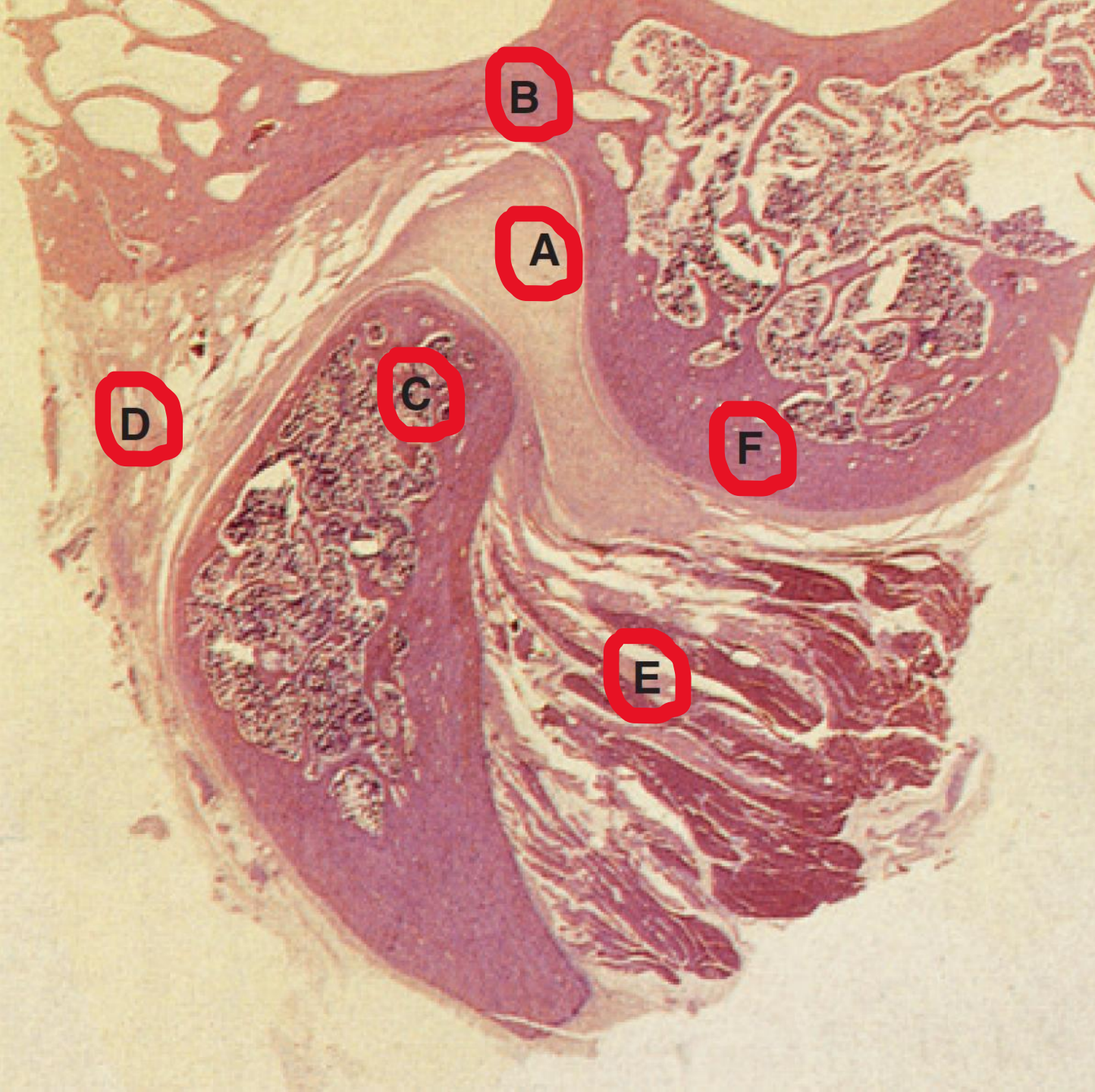
## Box 15.2 Features associated with cartilage

- the presence of type II collagen **Only traces** ✓
- the presence of high-molecular-weight chondroitin-sulphate-containing proteoglycans ✓
- the presence of a pericellular matrix surrounding the cells ✗
- chondrocyte-like cells that exhibit a rounded morphology with short, microvillus-like processes **Not true chondrocytes** ✓
- the absence of cell contacts ✗
- the absence of blood vessels within the tissue ✓

Cells are often termed **fibrochondrocytes** due to **mixed fibroblast–chondrocyte features**.

---





**Fig. 15.1**

**Low-power sagittal section showing general distribution of tissues of the temporomandibular joint.**

A = intra-articular disc;

B = mandibular (glenoid) fossa;

C = condyle of mandible;

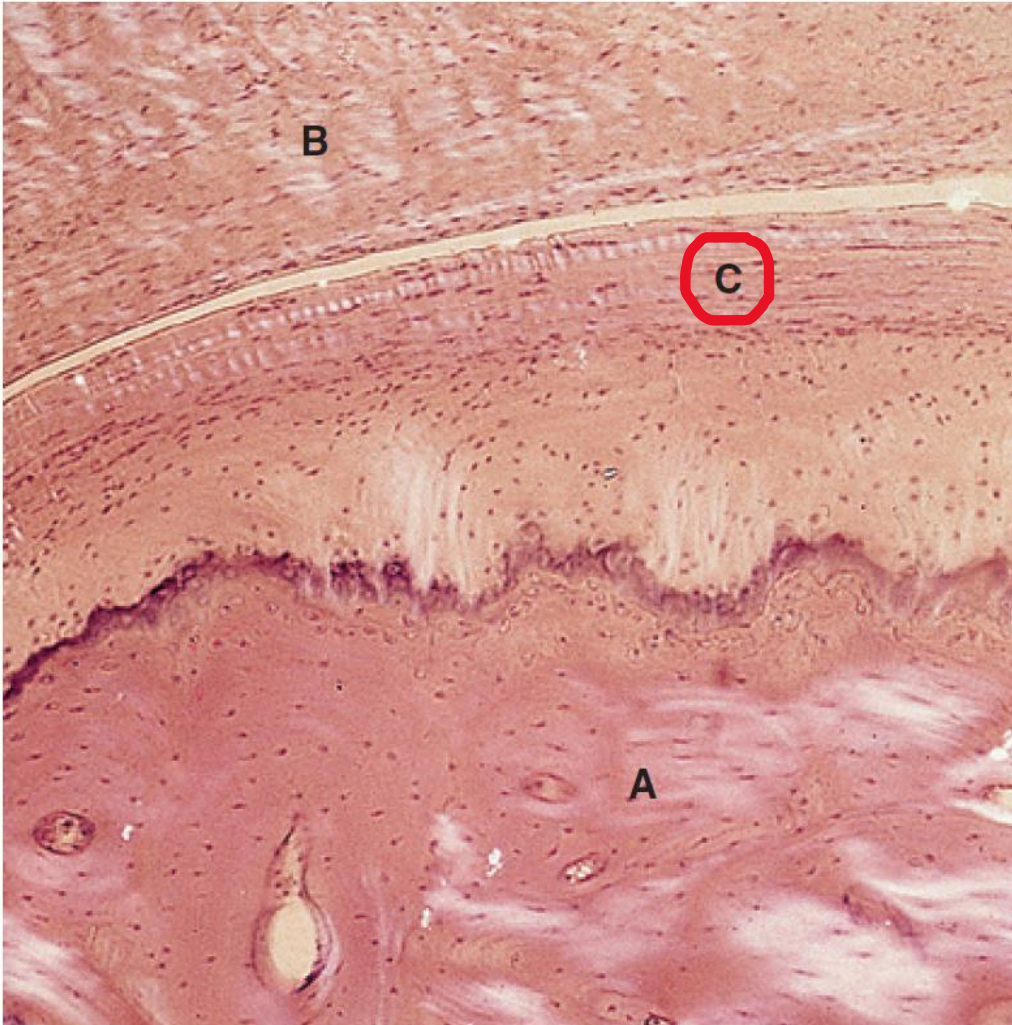
D = capsule of joint;

E = lateral pterygoid muscle;

F = articular eminence



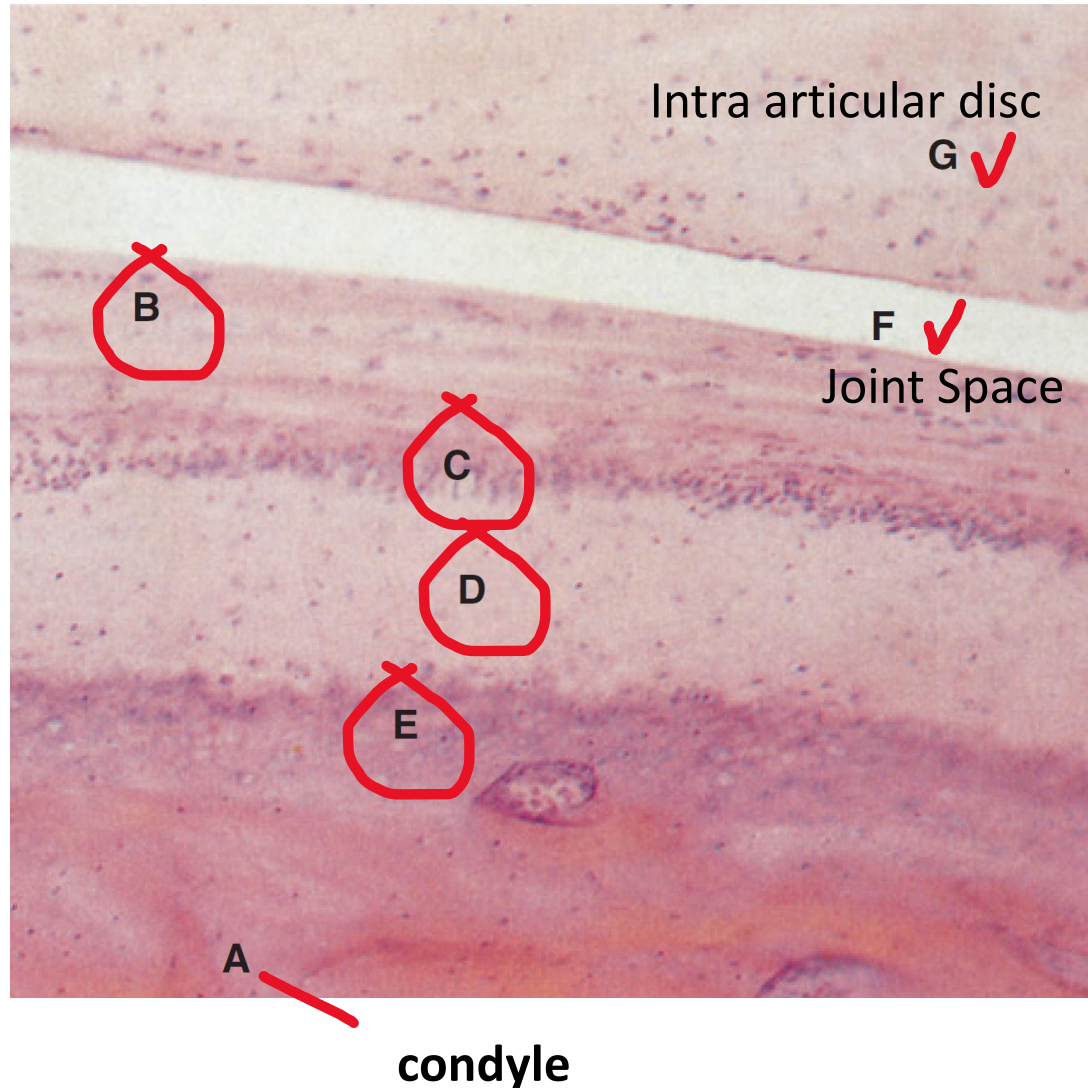
# Histological layers of the adult mandibular condyle



**Fig. 15.3 Sagittal section of head of adult condyle**

- A. Head of adult condyle
- B. Lower part of intra-articular disc
- C. Fibrous articular layer of the condyle

# Histological layers of the adult mandibular condyle



Four distinct histological layers:

## 1. Superficial Fibrous articular surface Layer; B:

Dense fibrous connective tissue, **collagen type 1**, some **elastin**, Sparse **fibroblasts/fibrocytes**.

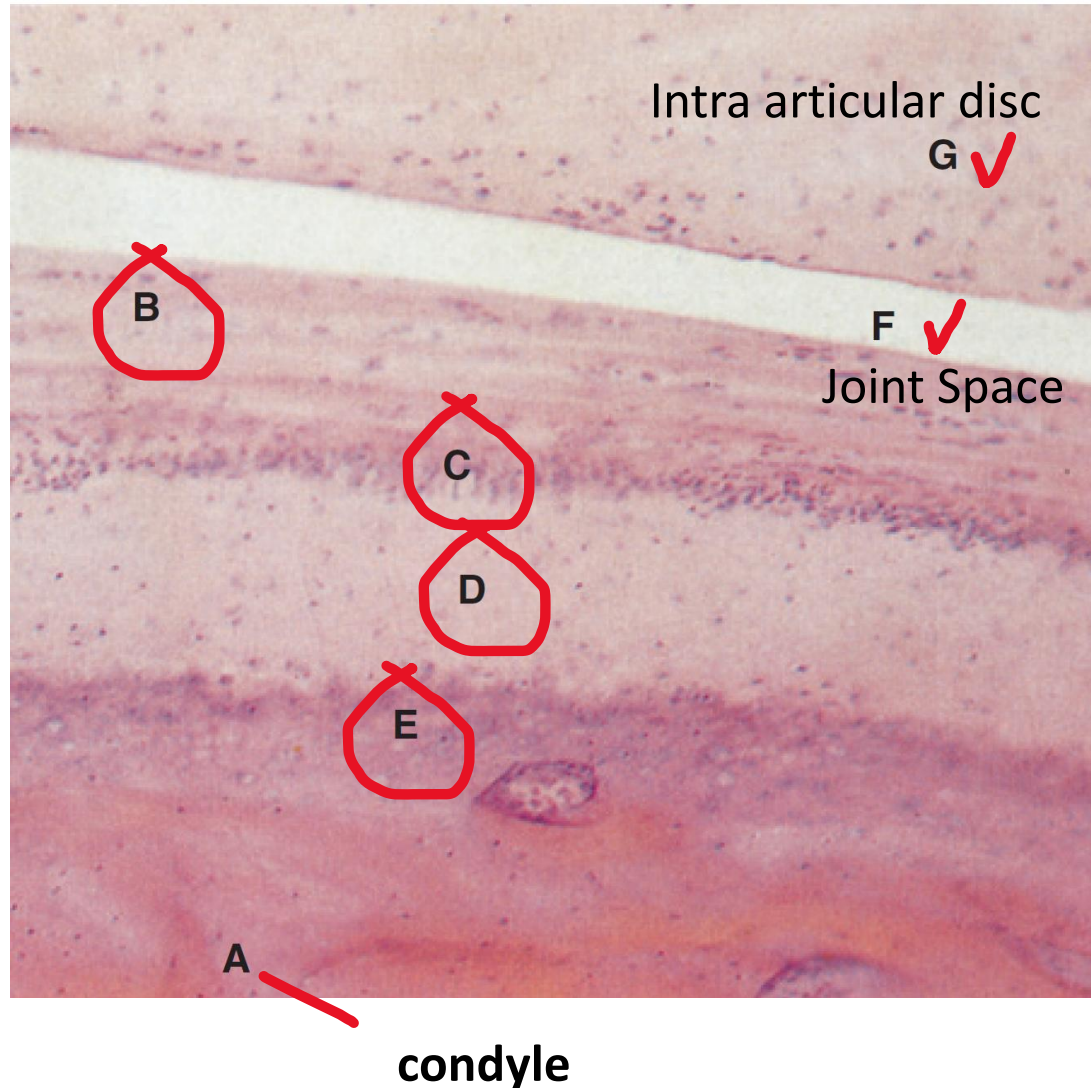
Collagen fibres **parallel** to surface. Show crimping. Resists shear. No chondrocytes here

## 2. Cell-Rich Zone: C:

Undifferentiated mesenchymal cells. Source of new cells for repair and growth. Cells can become fibroblasts or chondrocytes depending on mechanical stimulus.



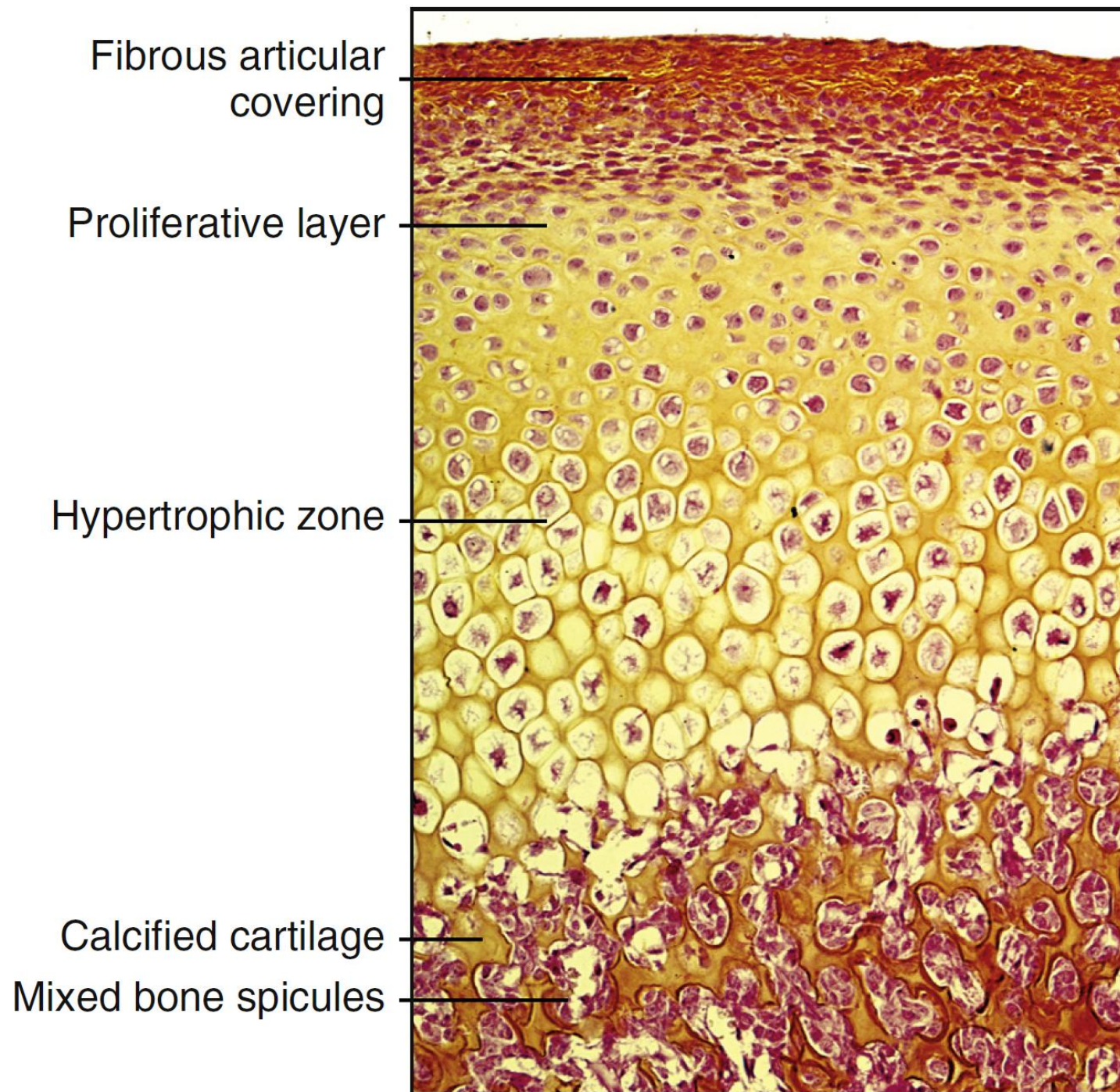
# Histological layers of the adult mandibular condyle



Four distinct histological layers:

**3. Fibrocartilaginous Layer: D:** Rounded, chondrocyte-like cells in dense CT matrix. Resembles fibrocartilage. Type I dominant + some Type II. Main functional layer in adults.

**4. Zone of Calcified Cartilage: E:** Remnant of secondary condylar cartilage. Just above subchondral bone. Distinct staining on H&E. "Tidemarks" can separate it from layer 3.



## Histological layers of the mandibular condyle **(Child)**

### **FIGURE 19-4**

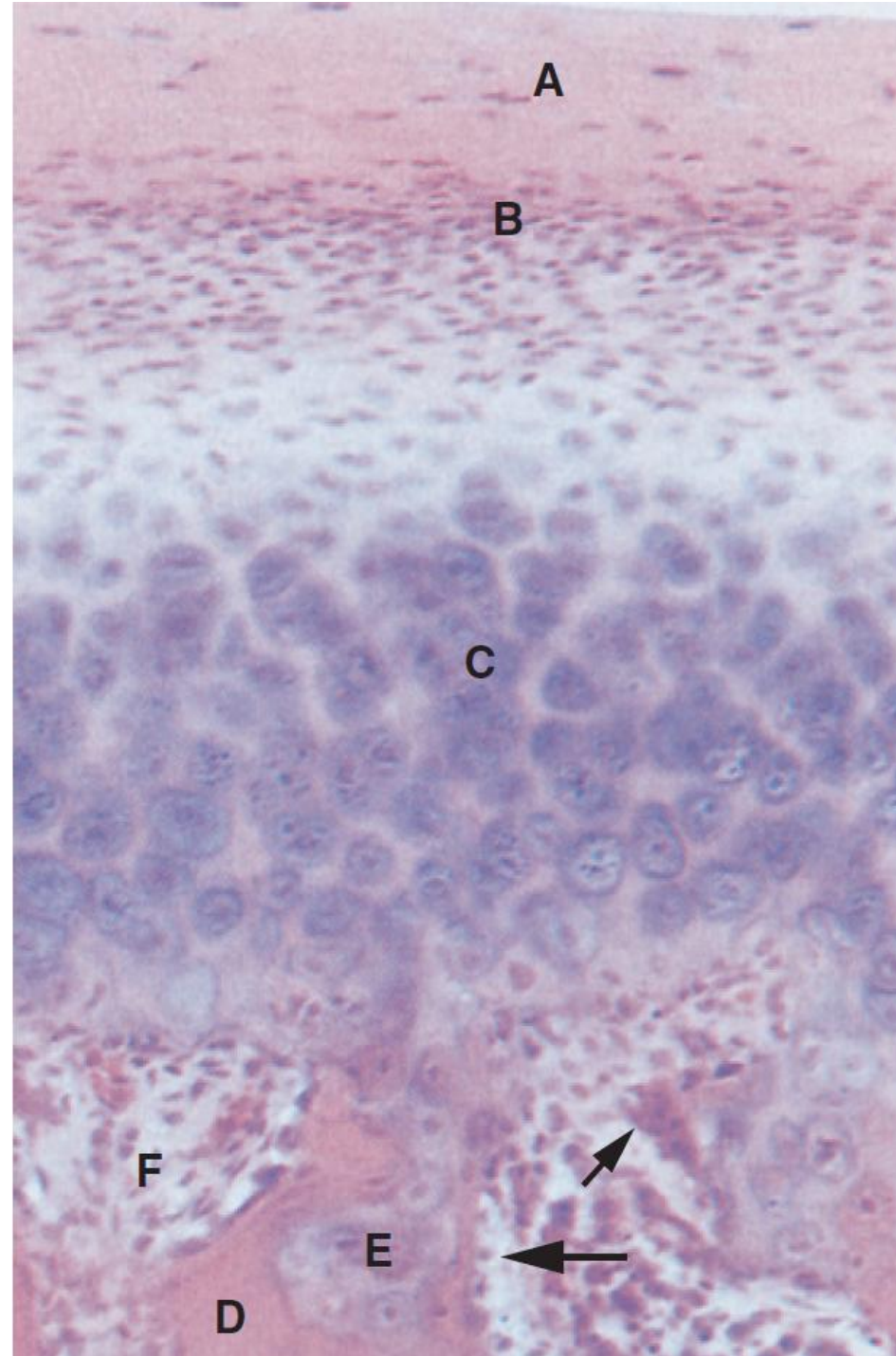
Growth center of mandibular condyle with endochondral transformation of hyaline cartilage into the alveolar process using interstitial growth. (From Nanci A: *Ten Cate's oral histology*.)



# Histological layers of the mandibular condyle (**Child**)

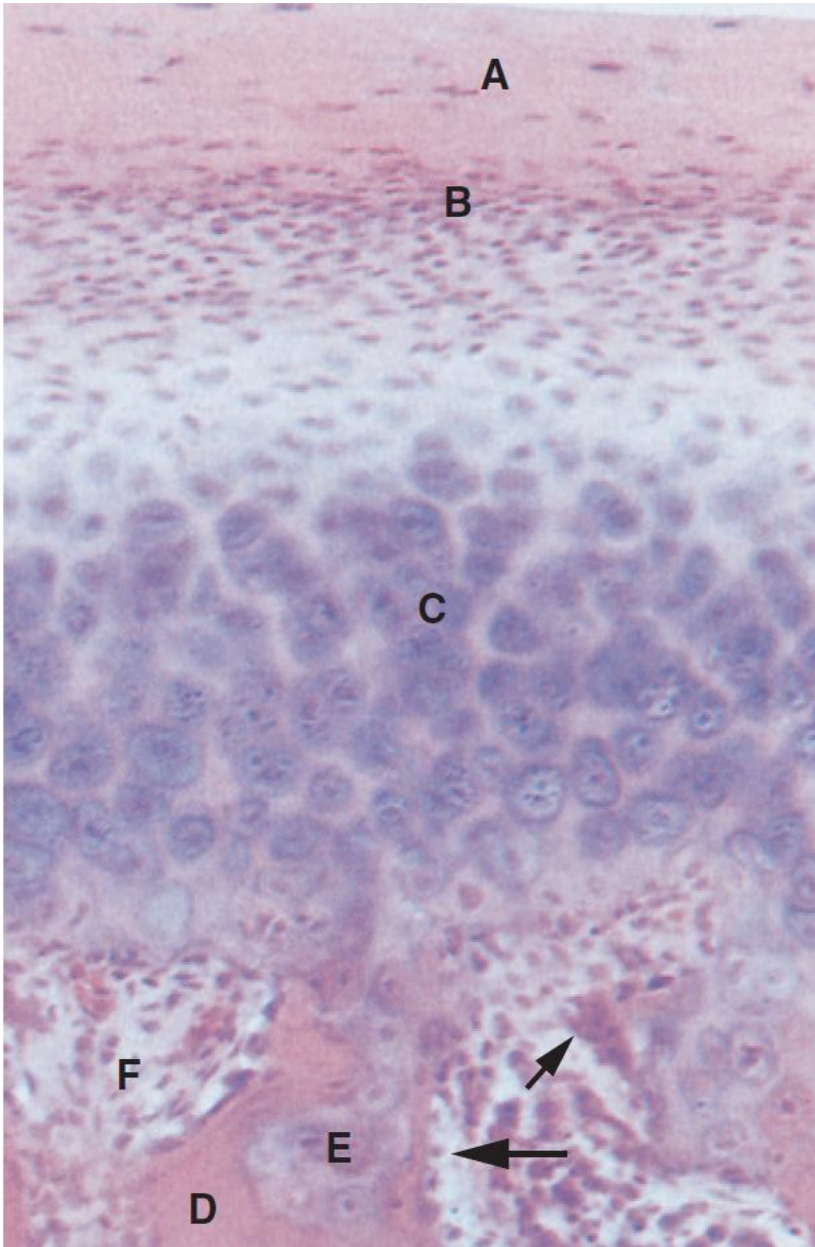
- **Four layers:**

- A; Outer fibrous articular layer;
- B; cell-rich proliferative layer
- C; Hypertrophic chondrocytes of the secondary condylar cartilage
- D; woven bone being deposited around
- E; a template of calcified cartilage
- F; marrow space



# Histological components of the mandibular condyle (**Child**)

- **Type I collagen** in all layers; **Type II** in layers 3 & 4; **Type X** in hypertrophic zone.
- **Chondrocytes** undergo hypertrophy → **endochondral ossification**.
- **No columnar alignment** of chondrocytes (unlike growth plate).
- **Osteoblasts** form bone on **calcified cartilage template**; **osteoclasts** resorb cartilage.
- **No compact bone lining** (unlike adult condyle).
- **Mechanical force** influences differentiation

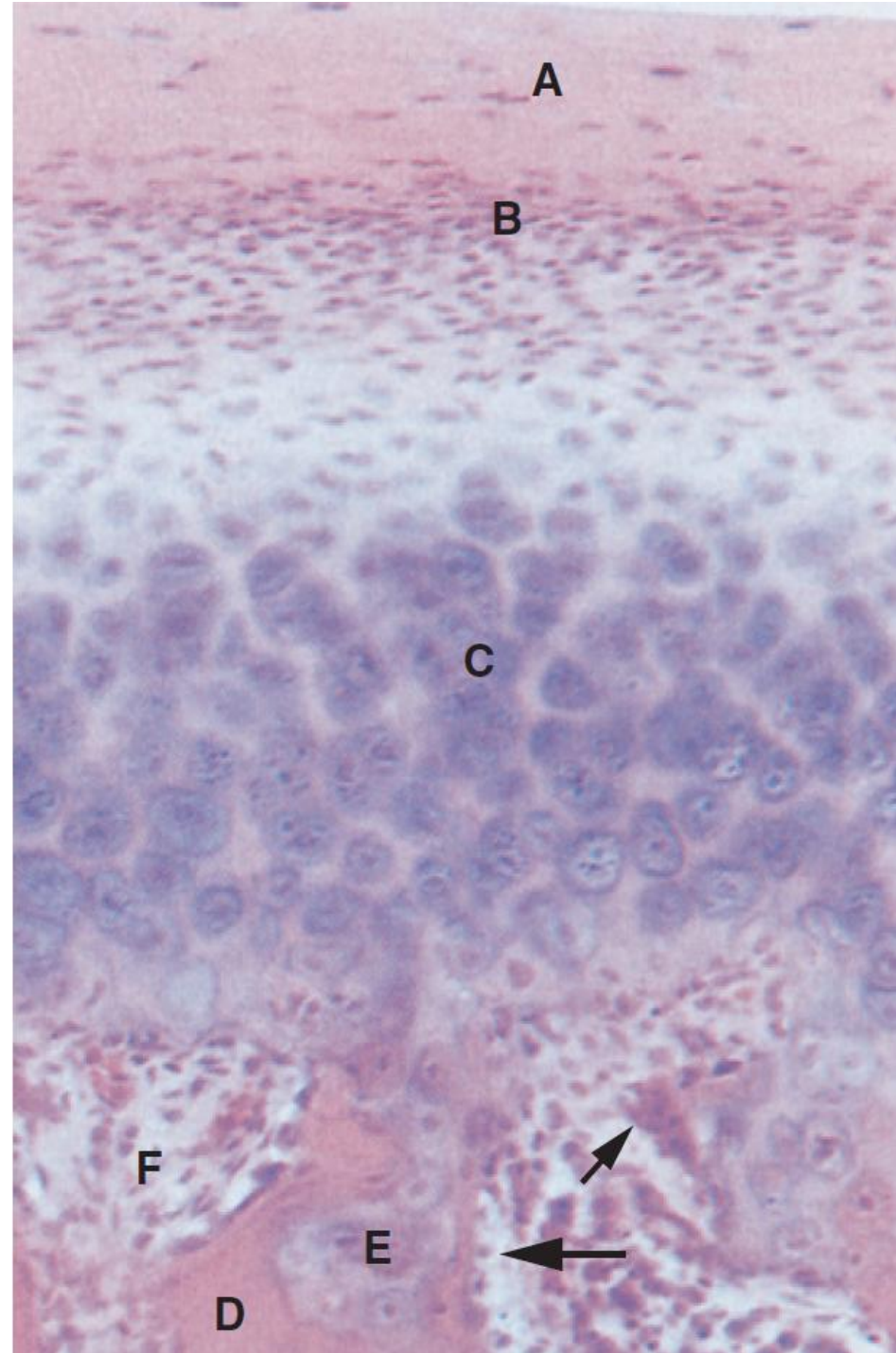




# Clinical Significance of the mandibular condyle (Child)

✓ **Secondary condylar cartilage** appears ~10th week IU, active until adolescence.

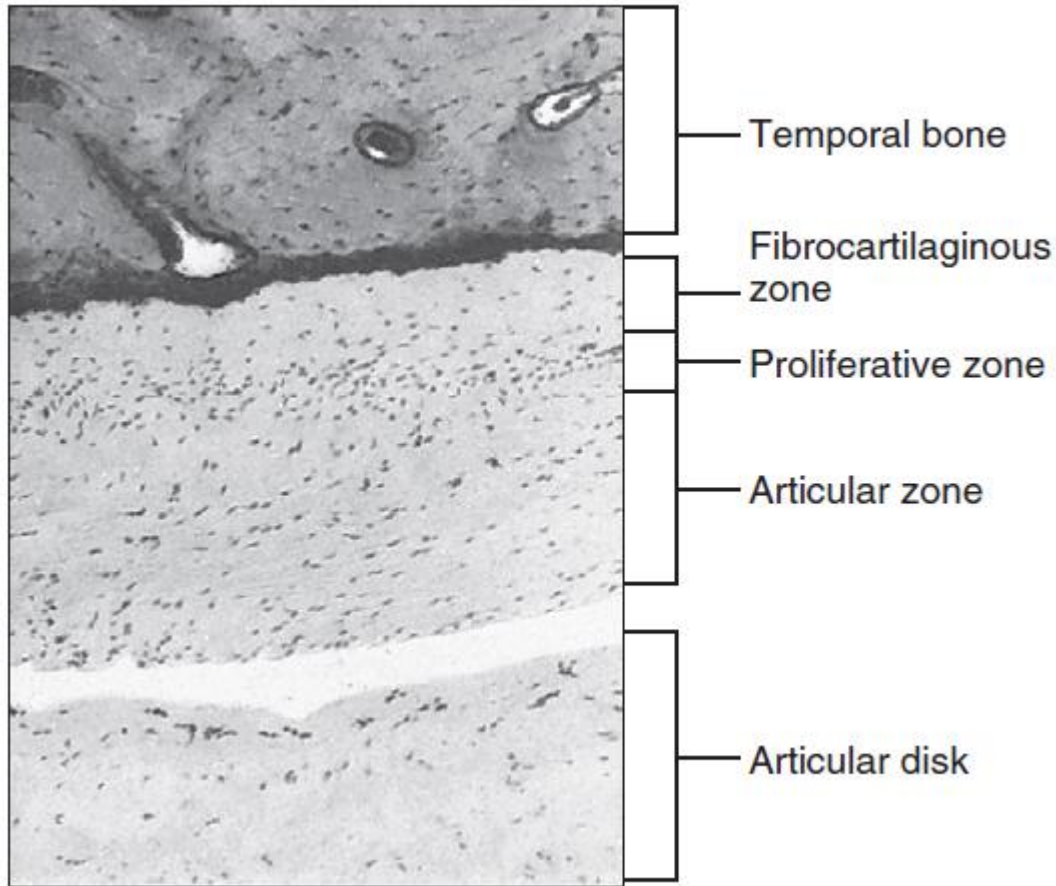
- It is NOT a primary(epiphyseal) growth cartilage — it responds to mechanical stimuli. That's why orthodontic/functional appliances in growing children and any trauma or injury to TMJ can influence condylar growth direction.





| Feature           | Child Condyle                                                          | Adult Condyle                                                  |
|-------------------|------------------------------------------------------------------------|----------------------------------------------------------------|
| Cartilage Type    | Secondary condylar cartilage (active)                                  | Replaced by bone; fibrocartilage on surface                    |
| Layers            | Fibrous → Proliferative → Hypertrophic chondrocytes → Calcified → Bone | Fibrous → Cell-rich → Fibrocartilage → Calcified bony spicules |
| Growth Activity   | Active endochondral ossification                                       | No growth; adaptive remodeling only                            |
| Collagen Types    | Types I (all layers), II (layers 3–4), X (hypertrophic zone)           | Primarily Type I; traces of II in layer 3, Type X is absent    |
| Bone Lining       | No distinct compact bone layer                                         | Compact bone lining present                                    |
| Cell Arrangement  | Present but NOT in columns (unlike growth plate)                       | No chondrogenic layers                                         |
| Response to Force | Mechanical loading drives differentiation                              | Adaptive remodelling (osteoclasts/blasts)                      |

# Histological layers of the mandibular fossa and articular eminence



- The articular surface of the mandibular fossa is similar in composition to that of the mandibular condyle.
- It is generally thinner than the condylar surface.
- Becomes thicker as it passes over the articular eminence, where functional loading increases.
- Superficial collagen fibers exhibit crimping (wavy appearance), which provides elasticity and resistance to shear forces during joint movement.

# TMJ Disorders – Key Points

- **Multifactorial:** Common signs include **pain, clicking, limited mouth opening.**
- **Normal TMJ:** Low friction due to **synovial fluid** and **smooth surfaces.**
- **Disorders:**
  - **Increased friction** with degraded synovial fluid
  - **Disc displacement** (commonly anteromedial) → **abnormal loading**





# High-yield summary

- TMJ articular surfaces covered by **fibrocartilage** (not hyaline) ✓
- Synovial membrane: Type A (phagocytic) + Type B(secretory) cells ✓
- Disc zones: PB (thickest) → IZ (thinnest, most collagen, load-bearing) → AB ✓
- IZ fibers: anteroposterior; Bands: transverse/oblique/vertical; Peripheral: circumferential ✓
- Disc main collagen: Type I (~80% dry weight) ✓

## High-yield summary

- Elastin: in bilaminar zone upper lamella + bands; absent in IZ ✓
- Disc cells = fibrochondrocytes (hybrid); gap junctions via Connexin 43 ✓
- Adult condyle: 4 layers (fibrous → cell-rich → fibrocartilage → calcified) ✓
- Child condyle: secondary cartilage, active endochondral ossification, Types I+II+X ✓
- Child chondrocytes: NOT in columns (unlike growth plate) ✓

## **Reference MCQ:**

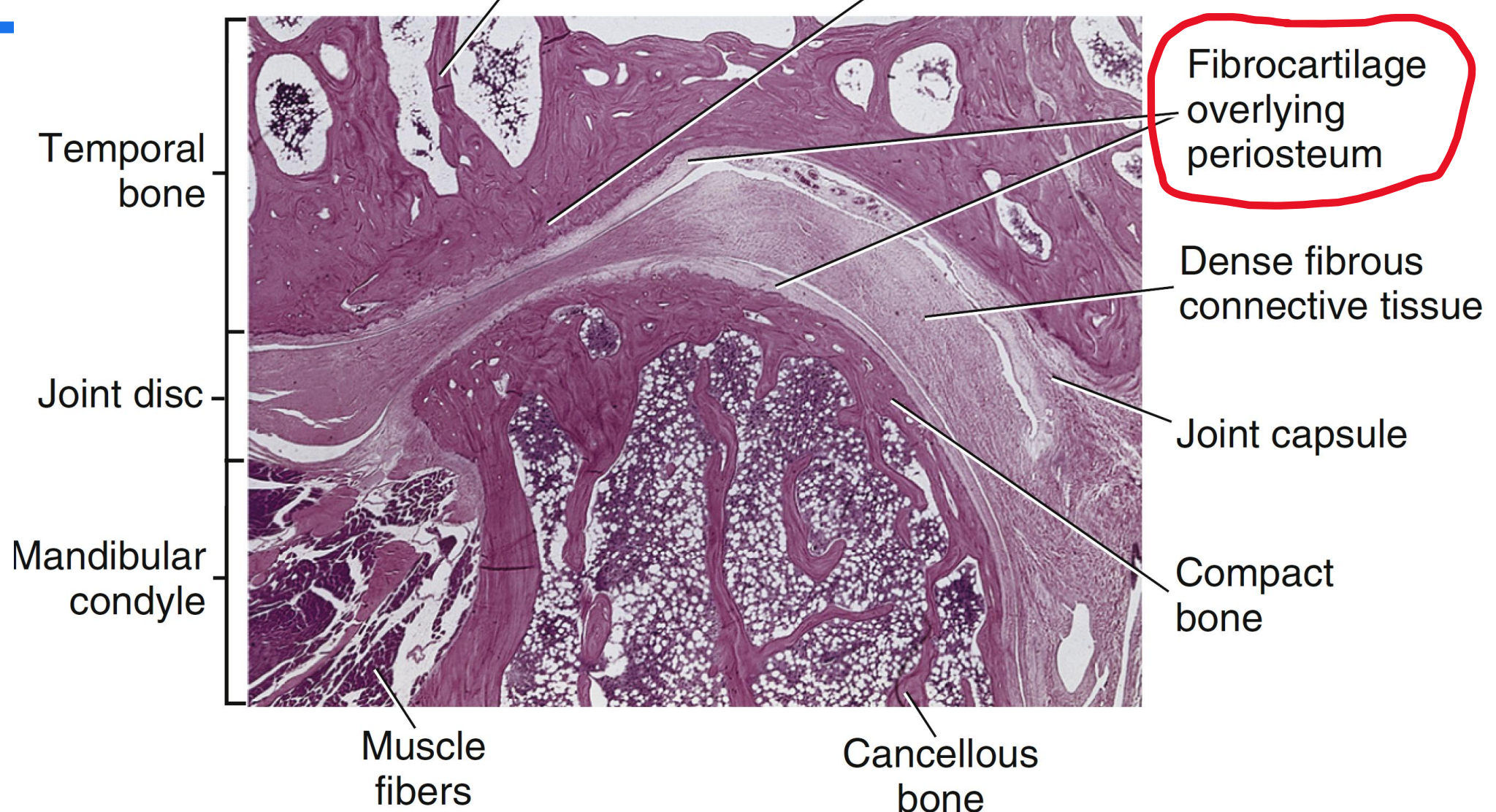
**The avascular TMJ disc receives nutrients necessary for survival. Which mechanism is primarily responsible for disc nutrition?**

- A. Active cellular transport
- B. Direct blood supply
- C. Diffusion and imbibition via synovial fluid
- D. Neural stimulation
- E. Lymphatic circulation

**Correct answer: C**



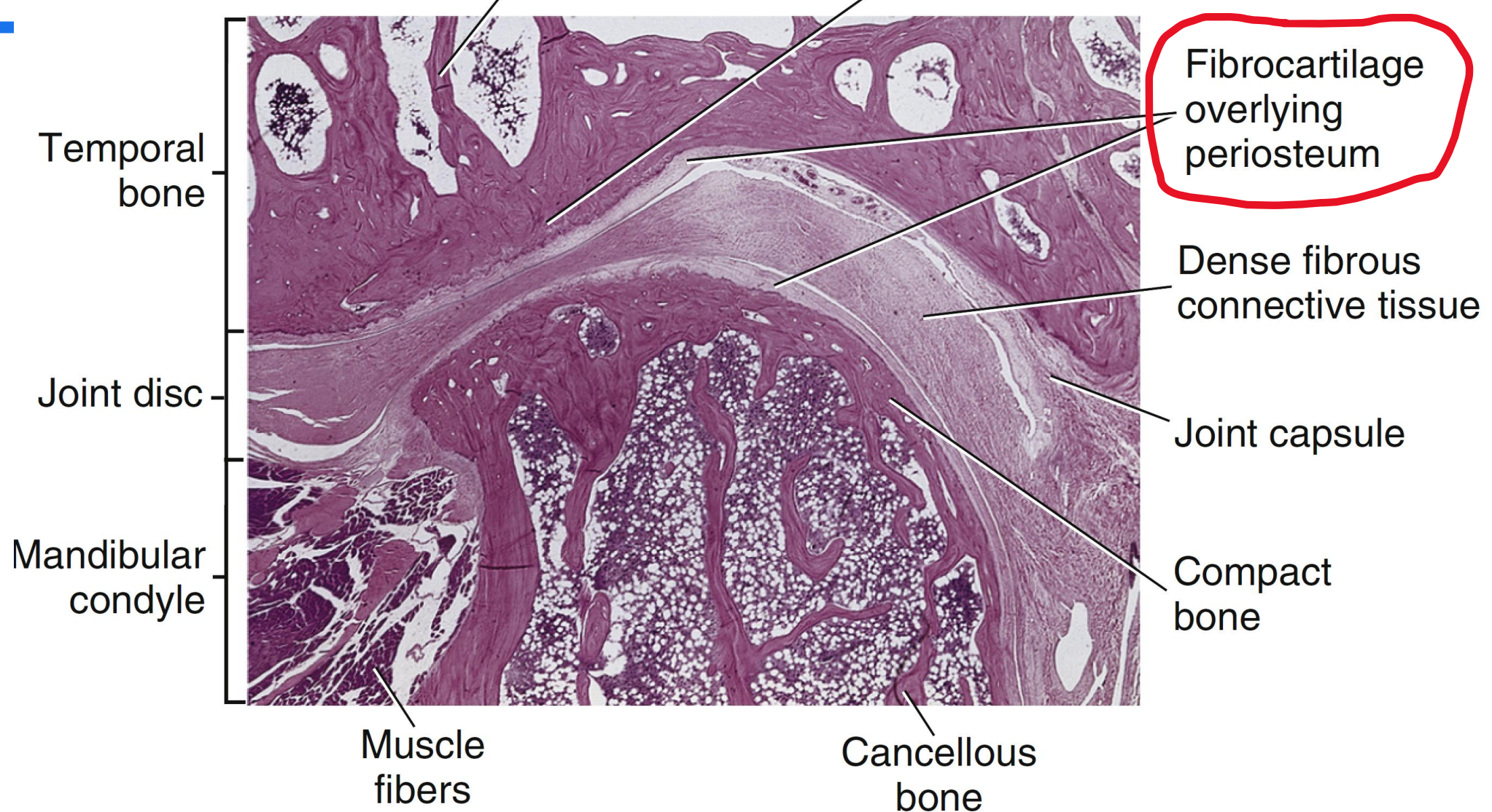
# Identify mandibular condyle, mandibular fossa and articular eminence



Articular Surfaces of the TMJ – Overview

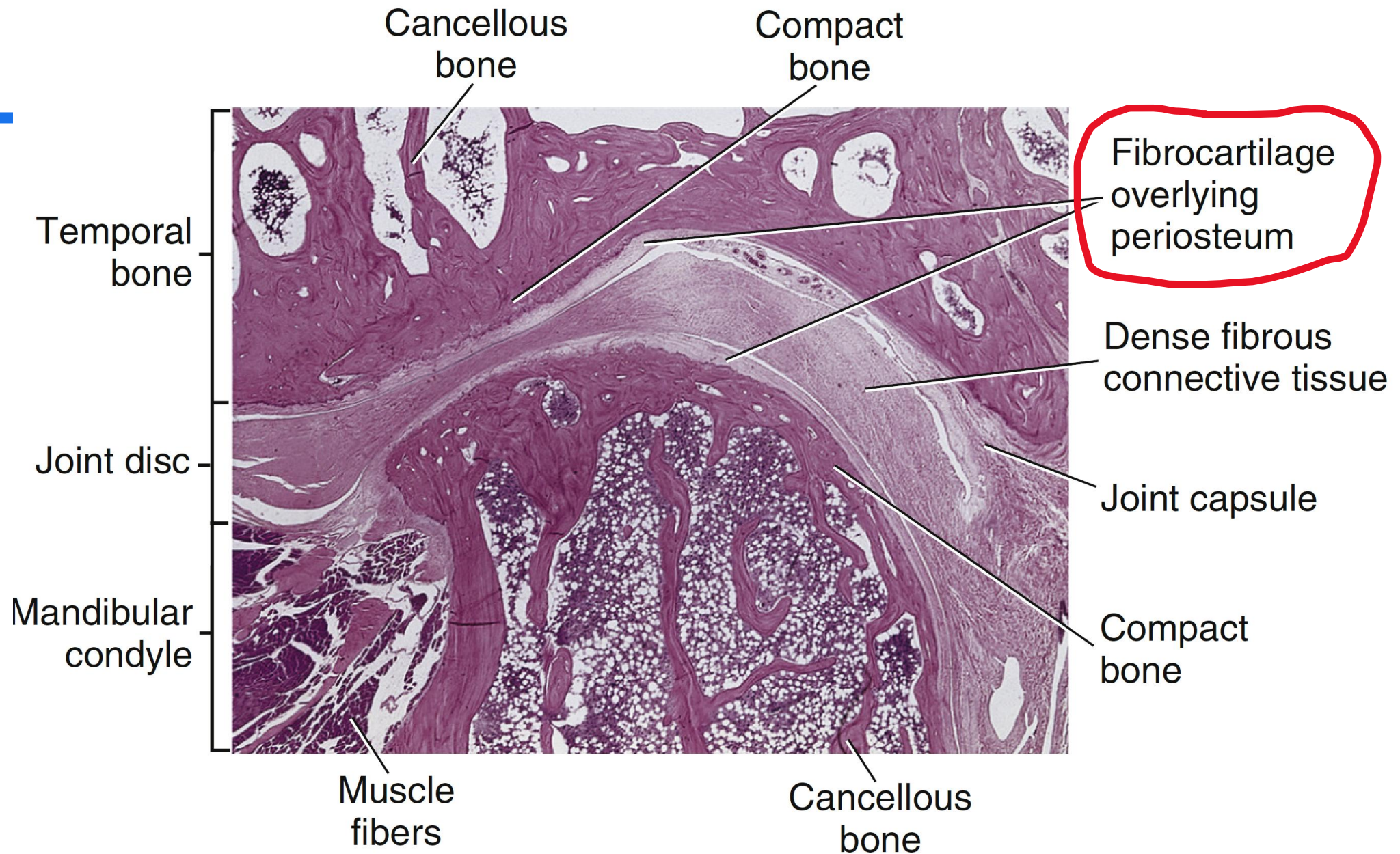


# Identify lateral pterygoid Muscle attachment



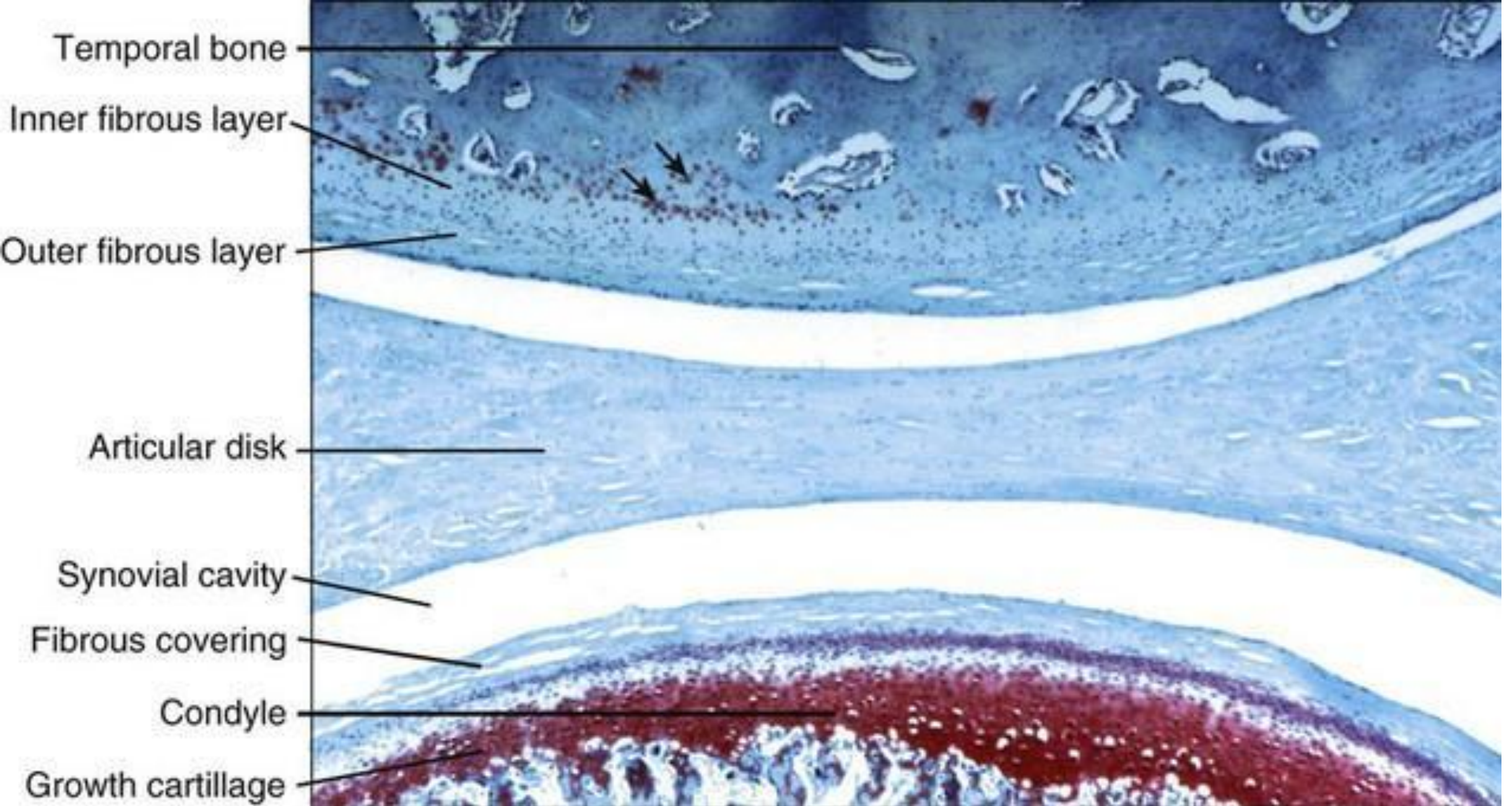
Articular Surfaces of the TMJ – Overview





Articular Surfaces of the TMJ – Overview





**Articular Surfaces of the TMJ – Overview**





Thank you  
Any questions

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